

VidiVerify

The 1-click check for your videos

User Manual

Desktop application for checking and repairing media files

Version 1.5

As of: October 2026

FREE, PRO and LIFETIME editions | For end users and technical professionals

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1. Overview and objectives

1.1 What is VidiVerify?

VidiVerify is a professional desktop application for checking and repairing media files, both video and audio. It checks files for errors, damage and completeness before they are archived, shared or put to productive use, one at a time or as an entire collection. Think of it as the vehicle safety inspection for digital media: a structured, reproducible and documented quality control. And when VidiVerify finds a defect, it can often fix it as well.

The application serves home users who want to check a media file now and then just as well as professionals in archiving, broadcast, post-production, video forensics, education and IT administration who need a reliable tool for daily work.

1.2 Typical problems

Digital media files can be damaged in many ways without it being obvious during playback. Common causes of hidden errors include:

- Aborted uploads or interrupted downloads
- Copy errors caused by unstable networks or faulty storage media
- Incomplete exports from video editing or conversion software
- Lost packets during streaming or recording
- Gradual corruption as storage media age and degrade
- Header structures that do not match the actual file content

> **[i] Why a dedicated check?**
> A file that opens and partly plays is not necessarily free of errors. Errors can hide in the last tenth of the file, individual frames can be damaged, or audio and video streams can drift out of sync. Only a complete technical check uncovers such problems reliably.

1.3 The core idea

VidiVerify brings together several established, high-performance, specialized plug-ins under one consistent, clearly structured user interface. These include FFmpeg and FFprobe for decoding and metadata analysis, MediaInfo for detailed media information, Magika for content-based file type detection and MKVToolNix for repairing Matroska containers. The in-depth image and frame analyses run on the VidiVerify runtime with the bundled professional libraries PyAV and OpenCV. Each plug-in contributes its own findings, and together they reveal more than any single tool could. The application

handles the complex parameterization automatically and presents the results in a clear, consistent form.

This makes professional file checking possible without in-depth knowledge of the underlying command-line tools. The proprietary MAP module (Media Analysis Protocol) summarizes all check results in a user-friendly view with clear color coding.

1.4 A concrete example: what VidiVerify takes off your hands

To illustrate the practical value, the following example shows the effort involved in checking a media file without VidiVerify, using command-line tools alone. A typical complete check would have to call several plug-ins in a coordinated way and then merge their output by hand, for example:

```
ffmpeg -hide_banner -v error -stats -xerror \
       -err_detect +explode+crccheck+bitstream+buffer \
       -fflags +discardcorrupt+genpts \
       -threads 4 -i "SampleVideo.mkv" \
       -map 0:v? -map 0:a? -map 0:s? \
       -c copy -f null - 2> check_log.log

ffprobe -v error -show_format -show_streams \
        -show_packets -show_frames \
        -of json "SampleVideo.mkv" > metadata.json

mediainfo --Output=XML "SampleVideo.mkv" > mediainfo.xml
```

This block of commands has to be typed by hand, adapted to each format, given the right paths and then interpreted. The resulting log files contain thousands of lines of raw information that still need to be evaluated. Error messages appear in technical jargon, individual notices are not sorted by relevance, and the result is plain text. In practice, a reliable assessment requires further plug-ins, such as MediaInfo for in-depth container data and Magika for content-based type checking, each with its own calls, parameters and output formats.

VidiVerify handles all of these steps automatically: you add the file by drag and drop, start the check with one click, and the results from the combined plug-ins appear in the MAP with color-coded ratings, ready to export as a protocol. Several minutes of manual command-line work become a single click.

1.5 Check levels and diagnostics

Three check levels build on one another and differ in depth and duration. In addition, the Diagnostic check mode extends any check level with an in-depth analysis. You can select it before starting, or VidiVerify offers it as soon as a check reports anomalies: the check escalates only where it is needed.

Check level	Duration	Purpose
Precheck	A few seconds	Automatically checks the file's accessibility, type and metadata when it is loaded
Integrity check	30 seconds (sample)	Decodes a defined part of the file and detects typical structural errors
Full check	Minutes, e.g. just over 8 minutes for a feature film (1:44 h, Full HD)	Decodes the entire file and also uncovers hidden errors
Diagnostic (check mode)	Sample of the first 30 seconds	Three depth stages (stream copy test, ffprobe deep analysis, decode layer) identify cause, severity and recommendation

PRO **Coming soon: VVProgressiv™**

VVProgressiv™ spreads short check windows across the entire runtime and goes into depth wherever something stands out. The result: a reliable verdict on the whole file in a fraction of the time a full check takes. VVProgressiv™ arrives by update and is included in PRO (Section 16.4).

1.6 VidiVerify in the age of artificial intelligence

Is VidiVerify still needed when AI can do everything? The straight answer: AI masters almost every individual step. It calls FFmpeg, interprets error messages and, on request, even writes its own check script that tests every file against the same scheme. And yet it does not replace VidiVerify. Building a scheme is easy. What counts is the knowledge inside it, the proof afterwards and the responsibility for the original.

- **Selected tools, one common standard:** VidiVerify makes specialized tools work together: FFmpeg decodes, MediaInfo reads the metadata, Magika identifies the true file type, MKVToolNix repairs containers. A dedicated, cross-media rating system translates their findings into one common traffic-light language, applying the same standard to video, audio and containers. An AI would have to reinvent this interplay for every question. In VidiVerify it is proven, versioned and documented.
- **A script is only as smart as its knowledge of errors:** The same image damage shows up in FFmpeg once as “mb incr damaged” and once as “concealing”. If you do not know that, you will not recognize the damage and will choose the wrong repair. Rules like these do not come from reasoning, but from real cases of damage, one after another. VidiVerify records them in its rule set and becomes more precise with every case it checks. An AI starts from scratch with every run.
- **Checking your own work is grading your own homework:** An AI that documents its own repair vouches for itself. VidiVerify independently records what happened: rule set, tool version, executed command, checksum before and after. That is the difference between a self-assessment and an inspection report.
- **Regenerated is not rescued:** For a family video, a concert recording or a digitized tape, there is no second source. An AI can only reinvent what is missing, and invented material is a new video, not the old one. Media Repair rescues what is there: never on the original, every result verified.
- **Private recordings stay private:** All checks run locally. No media file leaves your computer, nothing is uploaded to a cloud.

That is why VidiVerify is on its way to the next step: from a tool for people to a tool that AI assistants call on their own. The AI asks the question and explains the result, VidiVerify delivers the documented finding. Because the more answers an AI produces in seconds, the more valuable one question becomes: which of them can you trust?

> **[*] Our principle**
> The product is not the answer. It is the reason to trust it.

2. Key features at a glance

VidiVerify combines checking, rating, documenting and repairing media files in a single application. This overview shows the key features of version 1.5; features reserved for PRO carry the PRO badge.

2.1 File analysis and rating

- Automatic reading of container, video and audio streams, for video as well as pure audio files
- Display of resolution, aspect ratio, codec, bitrate, sample rate and bit depth
- Type integrity: detection of the true file type via the header signature (magic bytes) and the AI-powered Magika plug-in
- Rating of bitrate, resolution class and aspect ratio according to a dedicated, versioned rule set
- “Rating details & symbols” window: every rule and every threshold openly visible, with a classification of the loaded file
- Compatibility analysis in the Media Lab against six target profiles (Universal, Streaming, Smart TV, Mobile, Archive, Broadcast)
- Tools and advisors in the Media Lab: CodecLab, Bitrate EquivalenceLab and Video SizeLab

2.2 Check methods

- Precheck runs automatically when a file is loaded
- Integrity check with a 30-second sample
- Full check across the entire file
- Diagnostic check mode: three depth stages identify cause, severity and recommendation; when anomalies appear, VidiVerify offers the diagnostic on its own
- Strict abort on errors, optionally at the very first decoding error
- Media Batch: check entire folders and collections in one run, with pause, resume and StapelMAP (Chapter 7) PRO

2.3 Repair

- Media Repair: repair damaged files straight from the finding, with 16 methods, mandatory verification and RepairMAP as the audit sheet (Chapter 8) PRO
- Always works on a copy, the original stays untouched PRO

2.4 Documentation and proof

- The proprietary MAP (Media Analysis Protocol) as a designed A4 PDF archive sheet with clear color coding

- Every protocol states the rule set ID and the versions of the check tools
- A dedicated “Diagnostic deep analysis” section as soon as a diagnostic has run
- StapelMAP: check protocol for an entire batch PRO
- RepairMAP: audit sheet for every repair, with SHA-256 checksums of the source and target file PRO

2.5 Detailed analysis and diagnostics

- Diagnostic report in its own window: verdict, measurement tiles, one card per finding and all technical messages
- Detail windows for audio and video with level, spectrum, bitrate curve and waveform, plus detection of dropouts, inflated bitrates and fake FLAC files
- Read-only logs for runtime, exceptions and subprocesses
- Support package at the push of a button, optionally with privacy scrub (paths, host names, IP and MAC addresses masked)

2.6 Operation

- Modern interface in a consistent card-based design, from the main window to every dialog; all dialogs can be operated entirely by keyboard
- QuickCheck via right-click in Windows Explorer: a finding in seconds, without opening the main application
- “Open with” in Explorer for video and audio files
- Live telemetry with traffic-light colors: total system CPU, VidiVerify CPU, memory, network and read rate
- Three designs: Design mode, Windows classic, PLEX mode
- Five languages: German, English, Chinese, Spanish and Russian, matched to the system language on first launch
- Three CPU performance modes: Gentle, Balanced, High-Performance
- Drag and drop for one or many files
- Long operations keep the computer awake while the screen may still turn off

2.7 Updates and maintenance

- Self-update of the application, only with a verified checksum
- Separate update check for the plug-ins, downloads can be canceled at any time
- Every plug-in has a second download source, and VidiVerify switches to it automatically if the first fails or is slow
- Release notes as clearly arranged cards, even offline with the most recent version
- Storage & Maintenance automatically cleans up caches, backups and old runtime packages

2.8 Coming soon

VidiVerify keeps growing. These features are designed and will arrive by update; Chapter 16 describes them. Whatever is part of PRO is included in PRO at no extra cost.

- Media Compare: compare two versions of the same file side by side and keep the better one PRO
- Music Batch: check entire music collections for wrong bitrates, defects and duplicates PRO
- Perceptual quality analysis for image and sound PRO
- VVProgressiv™: distributed, adaptive checking across the entire runtime PRO
- MAP+ as a expert report PRO
- AI forensics, statistics module and YTUP™ PRO
- DVD and Blu-ray checks, NFO export for media managers, macOS, more languages
- VidiVerify as a check tool that AI assistants call on their own

3. License model: FREE, PRO and LIFETIME

VidiVerify comes in three editions. FREE is permanently free of charge for private use and a full-fledged check tool, not a demo. PRO adds the professional modules for entire file collections and for repairing damaged files; commercial use requires PRO. LIFETIME is PRO without limits: valid for all major versions, including future ones.

Feature	FREE	PRO and LIFETIME
Precheck, integrity check and full check	yes	yes
Diagnostic tool with diagnostic report	yes	yes
MAP as PDF archive sheet, “Rating details & symbols” window	yes	yes
Media Lab: compatibility analysis, CodecLab, Bitrate EquivalenceLab, Video SizeLab	yes	yes
Detail windows for audio and video, QuickCheck in Windows Explorer	yes	yes
Media Batch: entire folders in one run, pause and resume, StapelMAP	first file as a preview	yes
Media Repair: 16 repair methods, mandatory verification, RepairMAP	PRO notice	yes
Upcoming PRO modules of the current major version	no	included by update
Commercial use	no	yes

3.1 FREE: the full-fledged starting point

FREE includes all check levels, the diagnostic tool, the MAP and the complete Media Lab. It can be used without restriction for private purposes, for learning and for any non-commercial use, with no time limit and no watermark. All display designs, all five languages (German, English, Chinese, Spanish, Russian) and every setting are available in FREE as well.

FREE presents the PRO modules not as a locked door, but as a showcase: Media Batch checks the first file of a batch completely and presents the result, including the StapelMAP. This way, every user sees before buying what PRO delivers for their own collection.

3.2 PRO: VidiVerify at full capacity

PRO is designed for everyone who checks more than one file at a time or wants to rescue damaged files: commercial users, archives and organizations, and ambitious home users. PRO includes the

entire FREE scope and unlocks the professional modules immediately upon activation, with no reinstallation and no restart.

> **PRO What PRO gives you**
>
> PRO features are marked with a gold badge in this manual, for example:
>
> Media Batch PRO
>
> **In PRO today:**
>
> - Media Batch: check entire folders in one run, pause, resume later and document everything as a StapelMAP (Chapter 7)
> - Media Repair: repair damaged files straight from the finding, with a reasoned recommendation, mandatory verification and RepairMAP as proof (Chapter 8)
>
> **Coming soon, included by update:**
>
> - Media Compare: compare two versions of the same file side by side and keep the better one
> - Perceptual quality analysis for image and sound
> - MAP+ as a expert report
> - Further repair methods, for example to correct audio/video offset
>
> *More plans are described in the outlook (Chapter 16).*

One license for the entire major version

A PRO license applies to one major version, currently VidiVerify 1.x. Every extension added within this major version is included at no extra cost, and older major versions are covered as well. If you start today, you grow with the product up to the last release of the 1.x series. Before installing an update to a new major version, VidiVerify explicitly asks for confirmation, because PRO requires a new license there.

3.3 LIFETIME: license once, PRO for good

LIFETIME removes the major version limit. The license applies to all major versions, including future ones, with no subscription and no recurring costs. If you use VidiVerify for the long term, this secures the full future feature set. With PRO active, the purchase button in the license window reads “LIFETIME online upgrade”, and the purchase page then offers only LIFETIME, with the PRO you already paid for credited towards it.

3.4 Purchase and activation

All paths to PRO lead through the license window. Open it via “Help > Licensing …”.

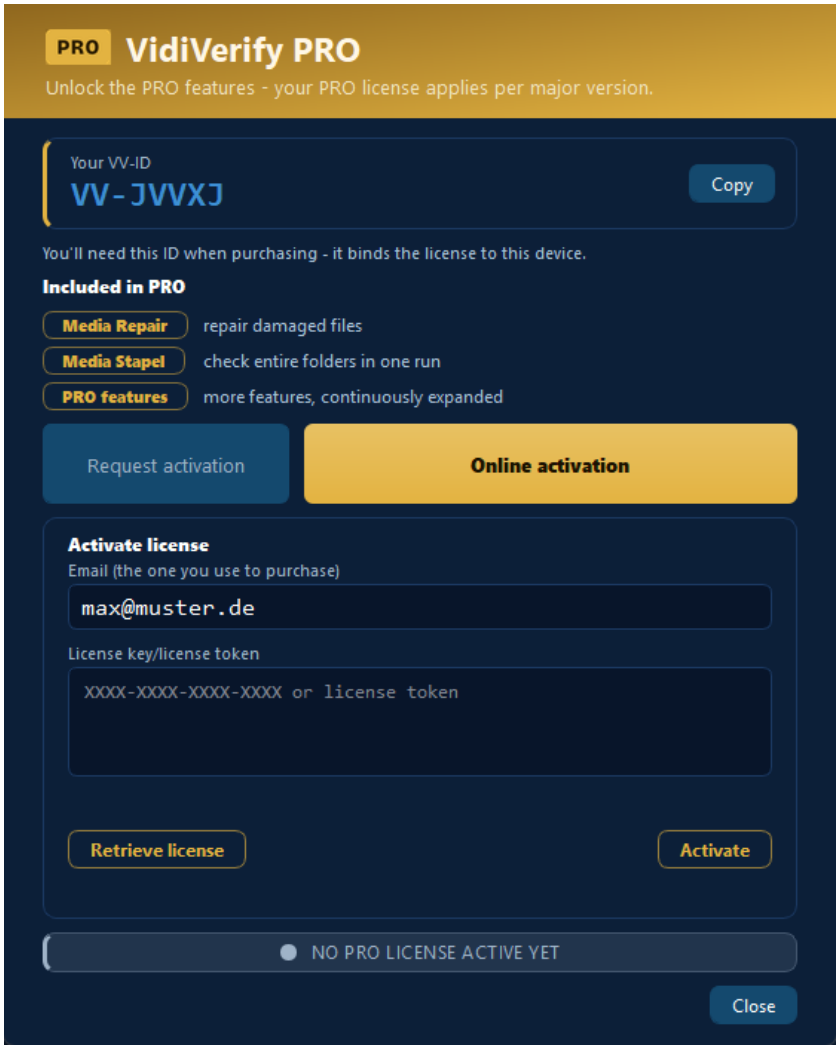


Figure 1: *The license window in FREE with both paths to PRO*

1. "Online activation" opens the purchase page on vidiverify.de. Your VV-ID and program version are passed along automatically, so there is nothing to type.
2. After payment, the license is created automatically. On the next launch, VidiVerify reports "PRO is now unlocked"; if you do not want to wait, click "Retrieve license" in the license window.
3. PRO is active immediately. VidiVerify downloads any required plug-ins, such as MKVToolNix, on its own.

For online purchases, the contracting party and invoicing party is Paddle as the merchant of record. VidiVerify itself processes no payment data and does not communicate with any payment provider.

If you prefer to buy by invoice, choose "Request activation". This opens a request form on vidiverify.de; the invoice and license key arrive by email. Click "Received a new key?", enter the key and confirm with "Activate".

> **[i] Tamper-proof**
> Every license is digitally signed (Ed25519). VidiVerify verifies the signature itself and rejects any altered or fabricated key.

3.5 Confirmation at launch

At every launch, VidiVerify confirms the license in the background without delaying startup. A renewed or extended license arrives automatically along the way. Without an internet connection, nothing changes at first: PRO stays active. Only when no confirmation has been possible for seven days does VidiVerify report “PRO is temporarily dormant” and continue in the FREE edition. The license remains stored and comes back to life with the next connection, or immediately via “Check now”.

The license status is shown on the splash screen, in the window title (for example “VidiVerify 1.5.0 (PRO)”) and in the About window.

3.6 Switching editions and keeping your data

You can switch from FREE to PRO or from PRO to LIFETIME at any time. Settings, protocols, saved batches and check results are fully preserved. If a license is revoked, for example after a refund, VidiVerify returns to the FREE scope, again without any data loss.

The Terms and Conditions, the right of withdrawal and the End User License Agreement (EULA) apply. They are available on vidiverify.de and in the app under “Help > License/EULA”.

4. Benefits for users and organizations

VidiVerify is designed as a tool that delivers compelling value both on a single workstation and across an organization. The following sections highlight the key benefits.

4.1 For end users

> **[+] Save time and avoid risks**
> Instead of reviewing media files manually or calling several individual tools, VidiVerify runs all relevant checks in a single pass. Errors are detected before a damaged file is archived or shared.

- A clear, guided workflow without a command line
- Meaningful, easy-to-read check protocols with color coding
- Instantly visible status indicators instead of cryptic error messages
- Three interface designs for different preferences
- Configurable CPU load, so the system stays responsive during a check

4.2 For technical professionals

- Full control over timeout, strict abort on errors and CPU settings
- Detailed event logs in the “Protocol” tab for diagnostics and traceability
- Clean separation between program directory and user data
- Freely accessible log files in plain text
- Versioned, openly viewable rule set as the basis for reproducible results
- Automatic compilation of a support package for submission to support

4.3 For organizations and archives

> **[*] Quality assurance as a documented process**
> Every check can be archived as a PDF, with the rule set and tool versions as proof. The result is a protocol for quality assurance, incoming inspection or forensic documentation. PRO adds the StapelMAP for entire collections and the RepairMAP for every repair.

- Audit-proof logging of check results
- Consistent check criteria across teams and locations
- Open, traceable storage of settings and logs
- No cloud dependency: checks run exclusively on your local machine
- Stable device ID for identification in support cases

- Media Batch: check entire collections consistently and document them with the StapelMAP PRO
- Media Repair: rescue damaged files, with an audit sheet for every repair PRO

4.4 Economic benefits

Aspect	Benefit
Less investigation effort	Faulty files are detected early, before time is invested in downstream steps
Avoiding data loss	Early detection of damaged files saves later recovery costs; Media Repair can rescue many of them
Fewer manual checks	Automated check runs replace spot-checking individual files
Consistent check standard	All reviewers use the same criteria and protocol structures
Clear license model	FREE for private use, PRO for professional use, LIFETIME for all major versions

5. Installation and setup

5.1 System requirements

Component	Recommendation
Operating system	Windows 10 or Windows 11 (64-bit), macOS planned
Memory	At least 4 GB, 8 GB recommended
Disk space	At least 200 MB for the application, plus space for plug-ins
Processor	64-bit processor with at least two cores
Screen resolution	At least 1105 × 753 pixels
Permissions	Standard user rights are sufficient; administrator rights are needed only for the initial installation

5.2 Installation steps

1. Download and run the installer.
2. Choose a language: setup speaks all five languages of the app and follows the Windows light or dark mode.
3. Confirm the target folder (typically C:\Program Files\VidiVerify).
4. Run the installation and start the application.
5. On first launch, the application automatically runs a plug-in check.
6. If plug-ins are missing, a guided download wizard is offered.

> **[+] Simple and safe**
> The setup wizard guides you through the entire process. The required plug-ins can be downloaded right away or later from within the application.

5.3 First launch and plug-in check

On first launch, a splash screen appears briefly. During this time, the application checks that all required plug-ins are available. If any are missing, an installation dialog opens automatically. You can start the download immediately or skip it; in that case, some check functions become available only once the plug-ins have been installed.

> **[i] Two sources for every plug-in**
> If a plug-in's download source is unreachable or noticeably slow, VidiVerify automatically switches to its own fallback source and verifies every package against a fixed SHA-256 value. Downloads succeed even on a freshly set-up Windows system, because Windows itself handles certificate validation.

5.4 Language selection

VidiVerify speaks five languages. On first launch it adopts the Windows language; you can switch under “Menu > Settings > Display & Design”.

- German (default on German-language systems)
- English
- Chinese (中文)
- Spanish (Español)
- Russian (Русский)

More languages, including Hindi and Portuguese, are planned.

5.5 Licensing and activation

After installation, VidiVerify runs as FREE. Purchasing, retrieving and activating PRO or LIFETIME is handled via “Help > Licensing …” and described in Section 3.4; confirmation at startup is covered in Section 3.5. The splash screen already shows the license status.

5.6 Uninstalling

Uninstall VidiVerify with the included uninstaller (unins000.exe) in the program folder or via the Windows Control Panel. During an interactive uninstall, VidiVerify asks two yes/no questions, each defaulting to “No”, so nothing is deleted by accident:

- Remove the machine-wide plug-ins under %ProgramData%\VidiVerify? (With “No”, FFmpeg, MediaInfo, Magika and MKVToolNix are kept and do not need to be downloaded again after a reinstall.)
- Also remove your personal VidiVerify data in the user profile (%LocalAppData%\VidiVerify)? (With “No”, settings, preview cache, protocols, saved batches and the QuickCheck inbox are kept and picked up again by a later reinstall.)

This lets you either remove VidiVerify completely or simply replace the program without losing your configuration and plug-ins.

6. Usage, operation and workflows

6.1 Splash screen and main window

The splash screen shows every step of program startup, including “Checking license status ...”, and then fades smoothly into the main window. The main window follows a consistent card-based design: clean surfaces, a shared alignment, and the active check area framed in gold.

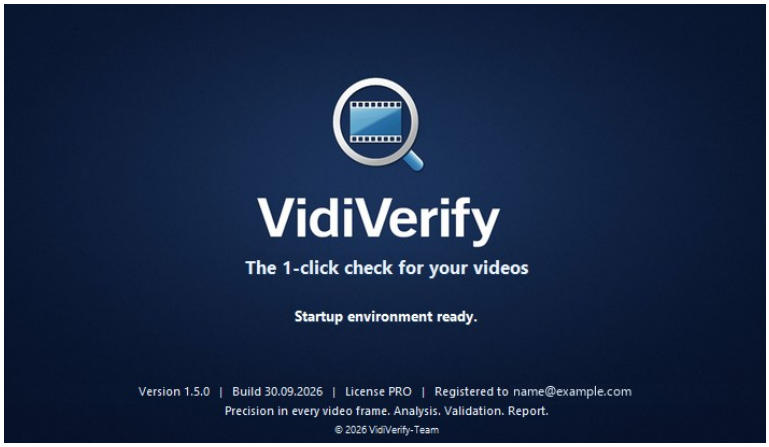


Figure 2: *The splash screen*

- At the top, the cards for media file, check level and check mode, CPU performance and threads, and live telemetry
- Live telemetry with traffic-light colors from 70, 80 and 90 %: total system CPU, VidiVerify CPU, memory, network and read rate
- The tabs: Overview, Media Analysis, Media Batch, Media Repair, Media Lab and Protocol; a gold band marks the selected tab, and the arrow keys switch between them
- The Overview with three cards: Preview, Media file and Check results; it grows with the window
- The preview with a thin gold frame; VidiVerify detects cover art, still image or image series on its own. Dolby Vision appears as “DOVI” with its profile, and for profile 5 with a notice that players without Dolby Vision support show incorrect colors

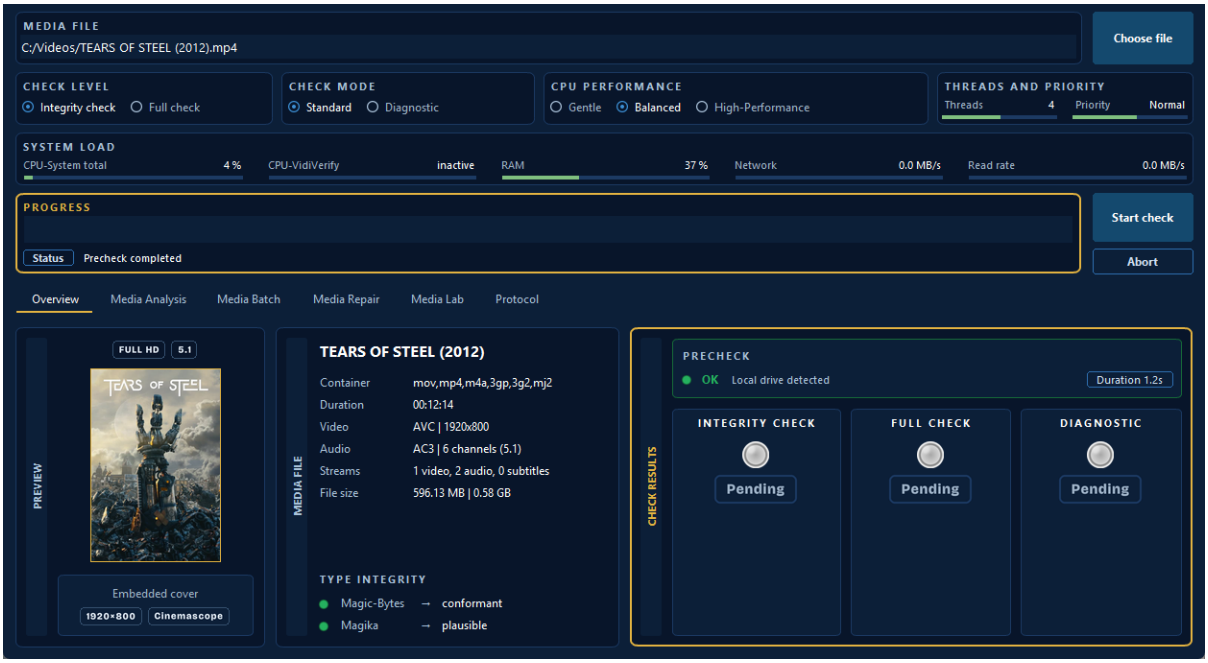


Figure 3: *The Overview after an integrity check*

6.2 Typical workflow

1. Start the application.
2. Choose a media file, drag and drop it, or hand it over from Explorer via right-click (“Open with” > VidiVerify).
3. Wait for the automatic precheck (a few seconds).
4. Choose the check level (integrity check or full check) and, if needed, the Diagnostic check mode.
5. Start the check.
6. Watch the progress or cancel if needed.
7. Read the result in the result cards and in the “Media Analysis” tab.
8. If needed, review details in the “Protocol” tab.
9. If anomalies are reported, click “Start diagnostic” (see 6.9).
10. Save the MAP as a PDF.
11. If a defect is found, repair the file with Media Repair (PRO, Chapter 8).

6.3 Precheck

The precheck starts automatically as soon as a file is selected. It checks basic accessibility and reads the metadata using the FFprobe plug-in. The result appears immediately in the information display. This stage delivers a first assessment within seconds, without fully decoding the file.

> **[i] Note on network paths**
> Files on network paths may need a longer precheck, depending on connection quality. The “Unstable network check” setting determines whether VidiVerify shows no warning, only warns, or blocks the check.

The precheck already points out known playback pitfalls that are explicitly not integrity errors. In these cases the file is technically intact; the notice concerns only playability in certain players:

- Fragmented streaming MP4 without an index (typical of IP camera and NVR recordings): some players cannot use the duration or seeking. The precheck card turns yellow with a short note; the finding also feeds into the diagnostic as an anomaly, with the recommendation to repackage the file once (remux).
- HEVC/H.265: a neutral notice with no traffic-light effect, stating that some players need an additional HEVC video extension for this codec format.

> **[i] Yellow does not mean “defective” here**
> A compatibility notice turns the precheck card yellow because something was detected and explained, not because the file is damaged. The actual check results (integrity check, full check and diagnostic) remain unaffected.

6.4 Integrity check

The integrity check decodes the first 30 seconds of the file. This section typically holds the container's most important header structures and index information. Errors here point to structural defects and are reliably detected. Runtime stays short, because only a defined excerpt is processed.

6.5 Full check

The full check decodes the entire file. This also reveals errors located in later parts of the file or affecting only individual frames. Runtime depends on file size, codec and computing power.

> **[+] In practice: a complete feature film**
> A feature film with a running time of 1:43:56, Full HD in AV1, 5.1 audio in E-AC3 and a size of 5.43 GB: the full check decoded every single frame in 8 minutes 11 seconds, in High-Performance mode with 8 threads. The integrity check (2.6 s) and the diagnostic (13.1 s) ran as well, all three without findings. How fast a full check runs depends mostly on the codec: the short film in the figure, 12 minutes in H.264, was fully checked in 12.6 seconds.

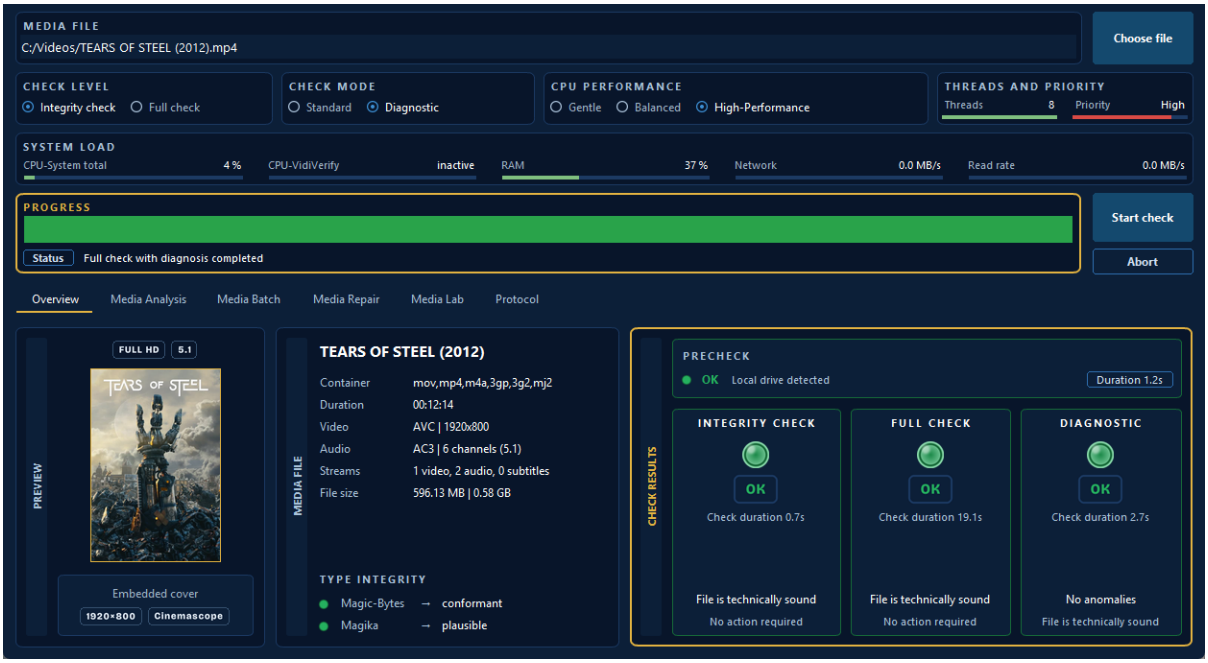


Figure 4: *All three checks passed, with their check durations*

6.6 Result cards, progress and cancellation

Each check level has its own card. The verdict appears as a badge inside it, and the border carries the traffic-light color of the result. While the check runs, the border is gold, and a point of light travels along the edge being worked on. The check duration sits on its own line; the explanation of the finding is in the tooltip.

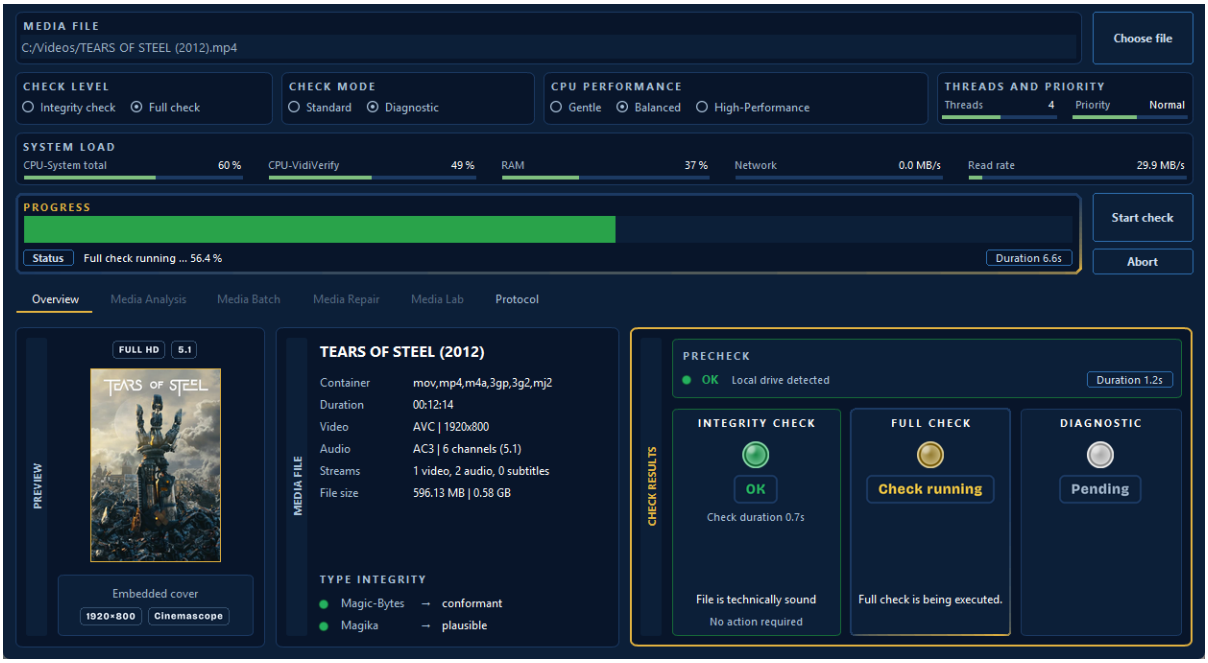


Figure 5: *Result cards during a running full check*

The progress display shows status and elapsed time. You can cancel a running check at any time; VidiVerify asks for confirmation first to prevent accidental cancellation. During long checks, the computer stays awake.

6.7 The “Protocol” tab

The “Protocol” tab shows detailed runtime information about internal processing: launched processes, timestamps, plug-in messages, detected stream information and technical notes. It is especially useful for technical professionals who want to trace the exact course of a check or classify anomalies.

The protocol consists of two cards: “Action” gives an overview of the most recent activity, either running or just completed, and “History” shows the complete, chronological chain of events. The current step is always visible without losing sight of the overall course. “Save protocol” writes the history to a file.

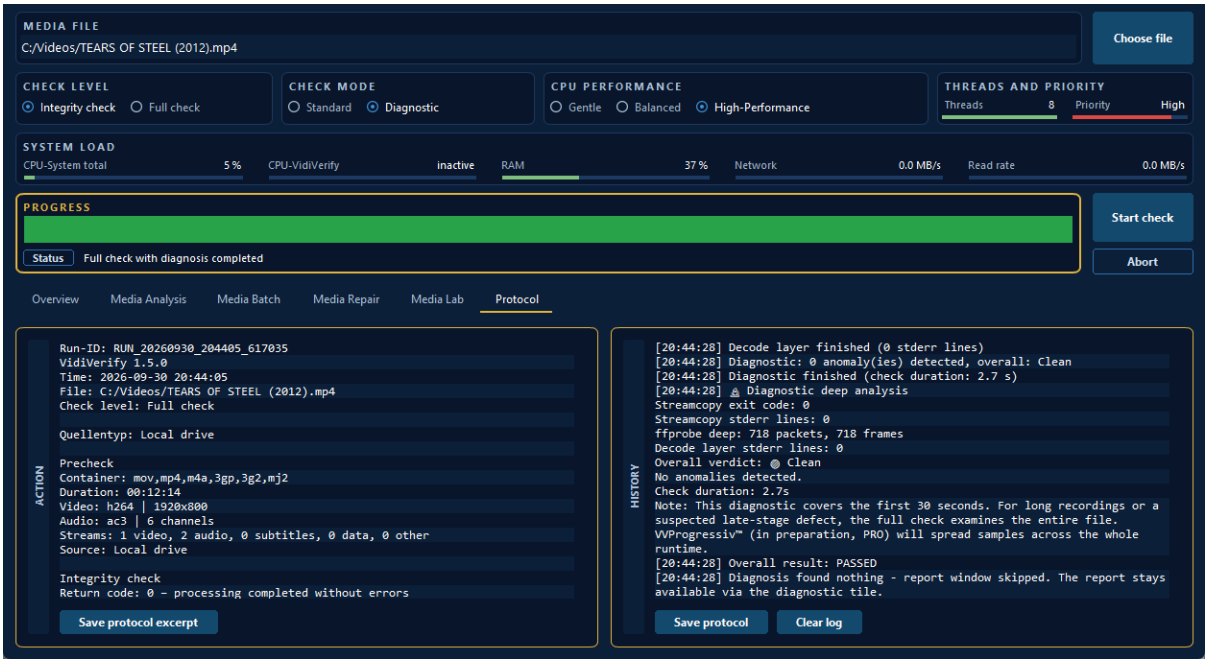


Figure 6: *The “Protocol” tab*

6.8 The “Media Analysis” tab

Media Analysis is VidiVerify's lab, and it is ready as soon as a file is loaded: right after the precheck, you can examine a media file here in full detail, without starting a check first. After a check, the tab also becomes the results page.

Two finding cards summarize video and audio: format, traffic light, condensed values and verdict. They open the detail windows, the real centerpiece (Section 6.11): playback with real-time frequency band and level, spectrum, cutoff, clipping, dropouts, keyframes, HDR data, sharpness and artifacts. What a technician would otherwise piece together from several specialized tools is right here in two windows.

Below them are three output cards:

- **MAP:** the check protocol as a PDF archive sheet (Chapter 10)
- **Rating details:** the rule set with all rules and thresholds, showing where the loaded file stands (Section 10.6)
- **MAP+:** the extended expert report, in preparation (PRO)

Without a readable data stream, the cards show dashes instead of invented values, and MAP output stays locked. For pure audio files, the video card stays empty. If a diagnostic was run, the MAP additionally contains the “Diagnostic deep analysis” section.

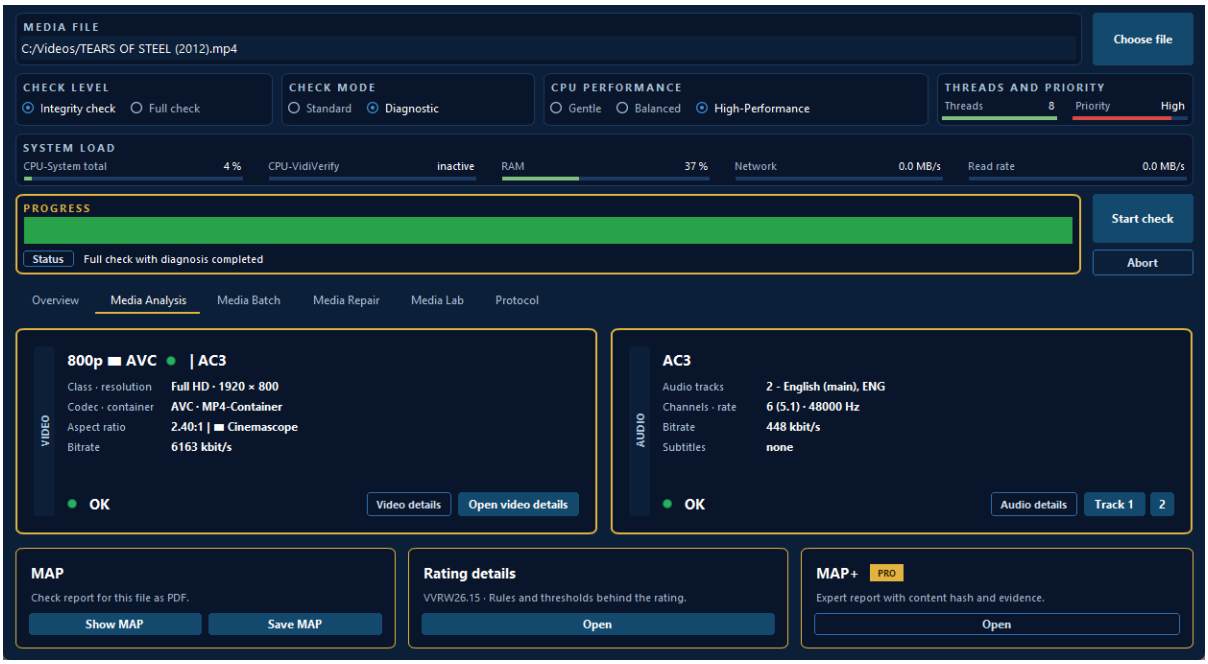


Figure 7: *The “Media Analysis” tab with finding and output cards*

6.9 Diagnostic tool

The diagnostic is the deep analysis for stubborn cases, selected as the “Diagnostic” check mode before starting or run afterwards via “Start diagnostic”. It examines the first 30 seconds of the file on three levels:

1. Stream copy test: repackages the container to verify structural consistency and uncovers timestamp and packet order problems.
2. ffprobe deep analysis: in-depth reading of packet and frame information, including timestamp, index and stream structures.
3. Decode layer: trial decoding that uncovers codec-specific defects not visible in the container.

From these three levels, VidiVerify derives a clear diagnosis: with cause, severity and a concrete recommendation. It reliably detects tape errors in VHS digitizations, defective DVD rips, aborted downloads, manipulated streams and damaged containers.

[+] Late defects

The diagnostic sees the first 30 seconds of a file. If the suspected problem lies further in, as with long recordings, the full check examines the entire file. VVProgressiv™ will distribute samples across the entire running time and dig deeper wherever something stands out: in preparation, included in PRO (Section 16.4).

[*] Understands quirks instead of raising false alarms

The diagnostic tool recognizes DVD cell transitions and H.264/HEVC B-frame structures for what they are: normal encoder behavior, not a defect. No more confusing red lights on flawless files.

Automatic escalation offer

If a check in Standard mode reports something unusual, VidiVerify offers “Start diagnostic” right away. One click is all it takes, and the diagnostic follows up.

The diagnostic report

The result appears in its own window: at the top, the verdict with a ring showing how many of the three stages passed without objection; below it, four measurement tiles, one card per finding, and all technical messages. The report opens on its own only when there is a finding; the diagnostic card opens it at any time. With PRO, “Start Media Repair” leads directly to the repair.

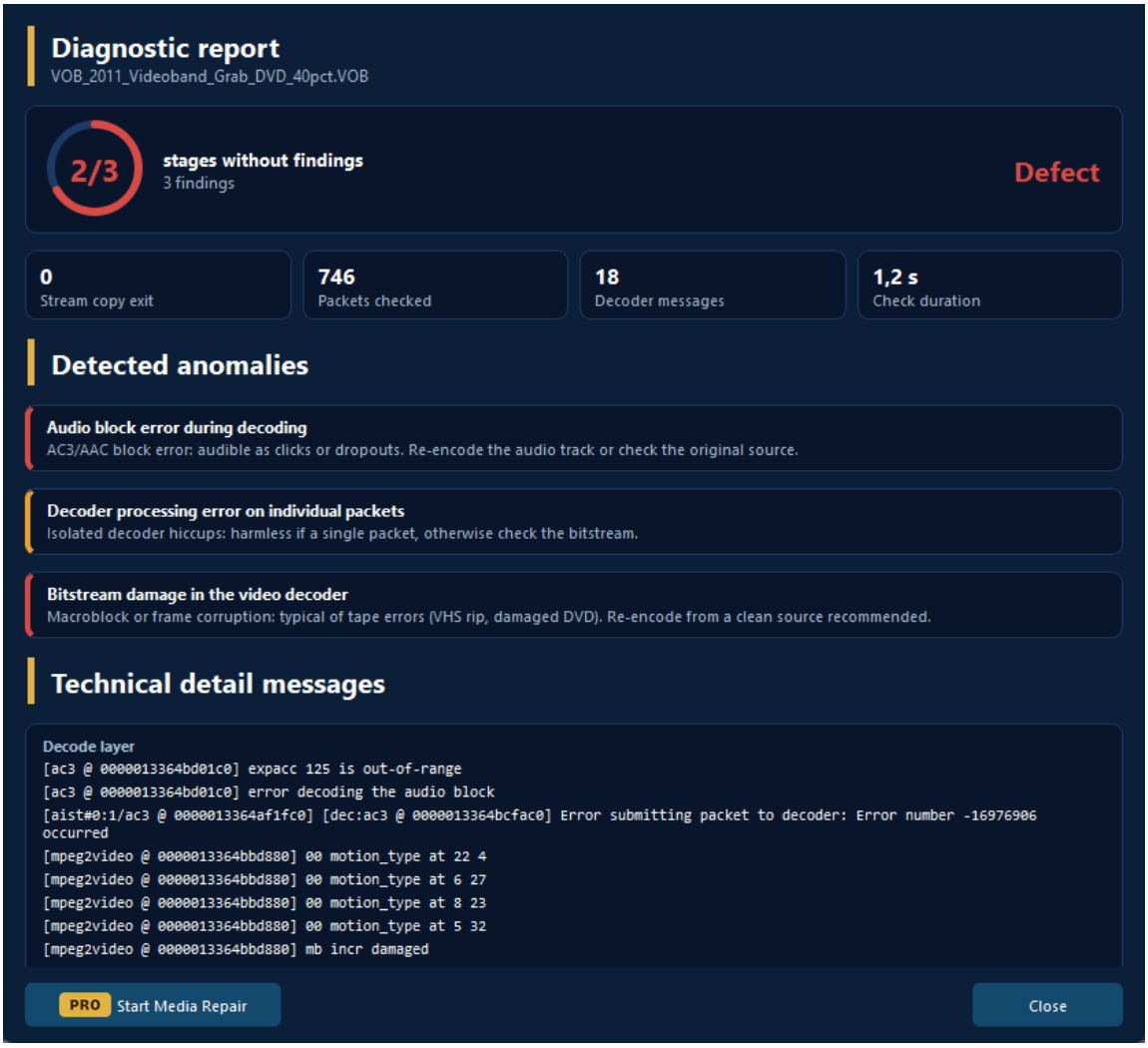


Figure 8: *The diagnostic report with a finding*

6.10 QuickCheck in Windows Explorer

QuickCheck is the fastest way to check a media file, without even opening the main application. Right-click a video or audio file in Windows Explorer and choose “Check with VidiVerify”: a lean, standalone worker component reads the container structure, the key media data and a first integrity finding within seconds and shows them in a compact notification window. The relevant core data is far quicker to take in than a native raw technical output.

- Available directly from the Windows 11 main context menu (top-level entry with its own icon)
- The notification window shows the status (OK, warning, error), a short finding and the central media data; several windows can be open side by side
- Optional hand-over to the full application: the result already determined is carried over, and the file is loaded there immediately
- For video files and all 22 supported audio formats
- Available in all five languages of the app

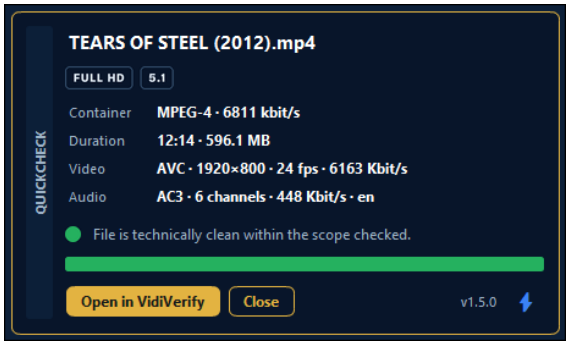


Figure 9: *The QuickCheck window*

> **[i] Also available: “Open with”**
>
> Independently of QuickCheck, you can also launch VidiVerify via “Right-click > Open with > VidiVerify”: the main application then opens with the file already loaded and starts the precheck immediately. Your default player stays exactly as it is: double-clicking a video file works as before, and VidiVerify simply appears as an additional option in the list.

6.11 Media Test: detail windows for audio and video

The two detail windows reveal what a file really contains. They open from the finding cards in the “Media Analysis” tab (“Open audio details”, “Open video details”), are part of FREE and are ready as soon as a prechecked file with readable data is loaded. Both play the file while the analysis runs along with it: you see and hear at the same time where a problem sits.

Audio details

- **Playback with level and frequency band:** During playback, VidiVerify shows level and frequency band in real time, with loop and mute
- **Frequency spectrum and spectrogram:** show how far the audio really extends
- **Bit-rate authenticity:** VidiVerify searches the spectrum for the edge that every lossy encoder leaves behind and compares it with the stated quality. The verdict reads “Plausible”, “Notable” or “Fake suspected”, together with the quality class that is actually in the signal, for example “sounds like 128 kbps”
- **Statistics:** RMS level, DC offset, zero crossings and clipping, meaning digital overload
- **Waveform and dropout detection:** level drops with a sharp edge are marked as strong findings, soft fluctuations as minor ones, directly in the timeline
- **Bitrate curve:** constant or variable, with average and range
- **Every audio track individually:** with multiple tracks, the main track is named and any other can be selected specifically

All charts can be zoomed and scrolled as in audio editing software.



Figure 10: *The audio details with spectrum and waveform*

[*] The encoder's fingerprint

Every MP3 or AAC encoder cuts off the treble at a fixed point: a hard level step with a flat shelf above it, in the same place for the entire runtime. This edge survives no matter how often the file is converted afterwards. VidiVerify measures it across the entire file and evaluates only loud passages. This exposes an upscaled 320 kbps MP3 just as reliably as a FLAC file that was made from an MP3. The method was validated against confirmed results from real collections of hundreds of MP3 and FLAC files.

Video details

- **Frame-accurate playback:** current frame, frame type, distance to the last keyframe, GOP length and remaining frames
- **Structure and bitrate:** bitrate curve with average and peak
- **Image and color:** pixel format, dynamic range and, for HDR material, mastering display and luminance
- **Perceptual analysis:** sharpness, noise, clipping, block artifacts and banding, each as current value and average

Without an audio output device, playback continues silently, and a warning card says so. The detail windows close with the main application and when a new file is loaded; VidiVerify asks first.

> **[*] The lab in two windows**
> Spectrum analysis, level metering, keyframe structure, HDR metadata and image quality: what would otherwise take several specialized programs, VidiVerify brings together in two windows, immediately after loading a file and already in FREE.

6.12 Checking pure audio files

VidiVerify treats pure audio files as a full-fledged category of their own, no longer as “video without picture” and certainly not as a supposedly defective file. When an audio file is loaded (Appendix C lists the supported formats), VidiVerify reliably recognizes it via magic bytes and the Magika plug-in and switches the interface to audio: cover display instead of video preview, audio-related key values in the information panel. It then runs through the same structured check levels as for video: precheck, integrity check and full check.

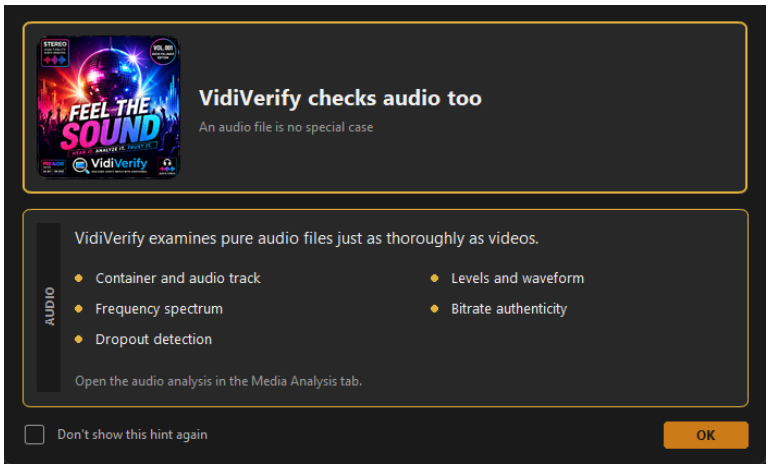


Figure 11: *The notice “VidiVerify checks audio too”*

The audio check shows its full strength in the audio detail window (“Media Analysis” tab → Audio details). It makes sound visible and uncovers problems you would otherwise notice only by listening very closely:

- Dropout detection: short audio dropouts, clicks and silent gaps are clearly marked in the level and waveform timeline, ideal for finding recording, editing or transmission errors before they end up in the finished product.
- Detection of inflated bitrates (upcoding/transcode): from the encoder edge in the spectrum, VidiVerify recognizes when a file is labeled with a high bitrate but the audio comes from a weaker source, such as an MP3 declared as 320 kbps that was upscaled from a 128 kbps file. The window also names the quality class that is actually in the signal.
- Fake FLAC files: a lossless format does not mean the recording was made without loss. If a FLAC file was made from an MP3, it carries that MP3's edge, and VidiVerify reports “Fake

suspected”. Genuine CD rips are recognized by the anti-aliasing filter of CD production and rated as plausible.

- Visual audio analysis: level curve, frequency spectrum, bitrate curve and waveform in one window, with zoomable, navigable charts as in audio editing software.
- FLAC with proof: the full check passes a FLAC file only if it decodes without errors and the embedded MD5 checksum over all samples matches.
- Correct classification even behind ID3 tags: FLAC and AAC with a prepended tag are no longer mistaken for MP3 or flagged as defective.

[*] Honest sound instead of false labels

For music and voice recordings, what counts is what is actually in the signal, not what the file header claims. VidiVerify checks both: the formal integrity of the file AND the real quality of the audio. Inflated bitrates, fake FLAC files and hidden dropouts are exposed before they become a problem.

7. Media Batch

An archive rarely consists of a single file. Media Batch checks entire folders, drives or any selection in one run, checking every file against the same rule set and with the same settings. The result is a filterable overview and, with the StapelMAP, a record for the entire collection. How many files in an archive are affected is no longer an estimate, but a finding.

> **PRO** **PRO feature**
>
> Media Batch is part of PRO. In FREE, you can try out the complete workflow (Section 7.6).

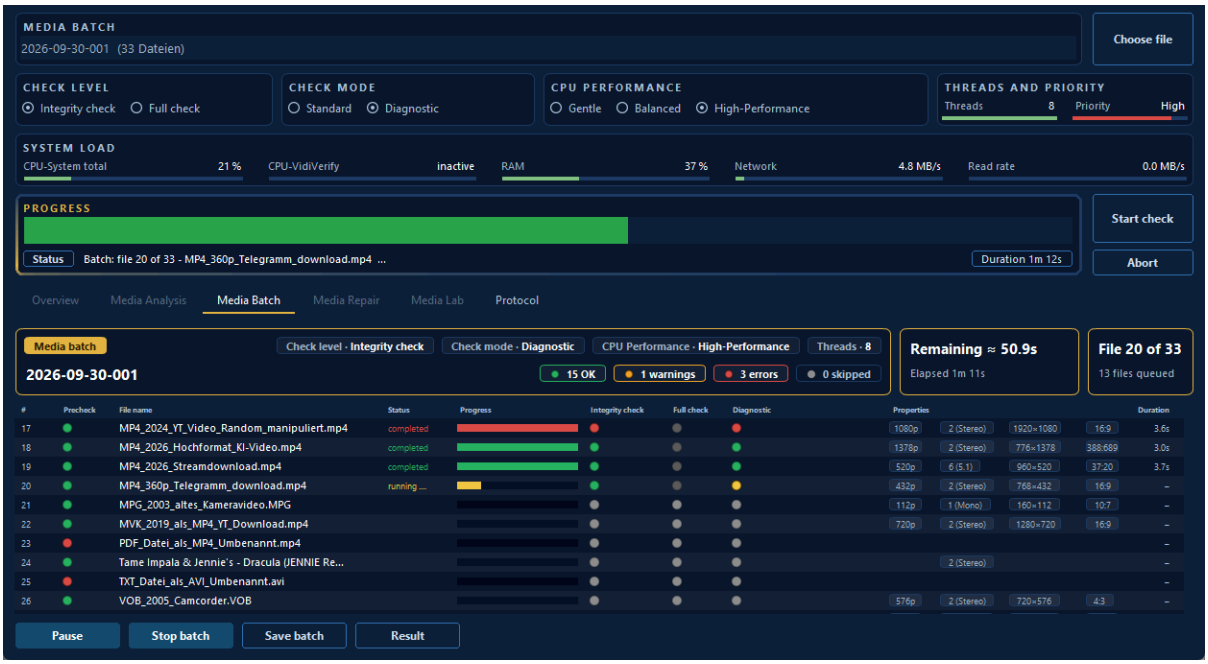


Figure 20: *Media Batch during a running check*

7.1 Creating a batch

On opening, VidiVerify asks “What would you like to check?” and offers these options:

- **Choose files:** Select one or more media files. If you select several, a batch is created.
- **Choose folder:** All media files in a folder. VidiVerify first states the number and asks whether to include subfolders: “This folder only” or “Include subfolders”. This prevents an accidental batch of thousands of files.
- **Resume batch …:** Appears as soon as open batches exist and picks up an interrupted batch (Section 7.3).

Multiple files dragged onto the main window can also be checked as a batch. VidiVerify then asks: “Check as a batch” or “Check one file only”.

First, the precheck runs over all files, visible in the main progress bar (“Pre-check · file 3/40”). Unreadable files thus stand out before the actual check. Files that are no longer at the specified

location are listed up front as “File not found”; sleeping network drives delay the process by no more than three seconds per folder. The check level and check mode from the main window apply to the entire batch. Click “Start check” to begin the run.

7.2 During the run

Each file appears as a row with file name, properties, progress and status; a tooltip for each check level shows the individual result. Above the list, you see elapsed and remaining time, the queue length and the running tally of OK, warnings, errors and skipped files.

- “Pause” halts the batch between two files, “Resume” continues it.
- “Stop batch” cleanly finishes the current file and then stops.
- During the run, VidiVerify prevents the computer from going to sleep; the display may turn off.

7.3 Pause, save and resume

After each checked file, VidiVerify saves the progress automatically. The batch therefore survives even a crash or closing the application. “Resume batch …” lists all open batches with their progress (“12/40 checked”) and resumes the selected one exactly where it left off. VidiVerify cleans up open batches older than 90 days on its own.

“Save batch” stores a batch as a file (extension .vva), and “Load batch” opens it again. When closing a batch, you choose between “Save”, “Discard” and “Keep open”.

7.4 One batch, one tool version

An assessment of a collection is only as valuable as its comparability. That is why every batch records the rule set and tool versions it is checked with. If anything has changed since the batch began, for example through an FFmpeg update, VidiVerify asks before continuing and offers exactly two options: “Re-check in full” or not continuing. A batch whose first half was checked under different rules than its second half simply cannot arise.

7.5 Result and StapelMAP

“Result” opens the “Batch Result” window. Key figure cards summarize the run: files, OK, warnings, errors and total duration. Clicking a card filters the list, for example to “Errors only”. For each file, a tooltip shows the complete check protocol; below the list are the check conditions of the run.

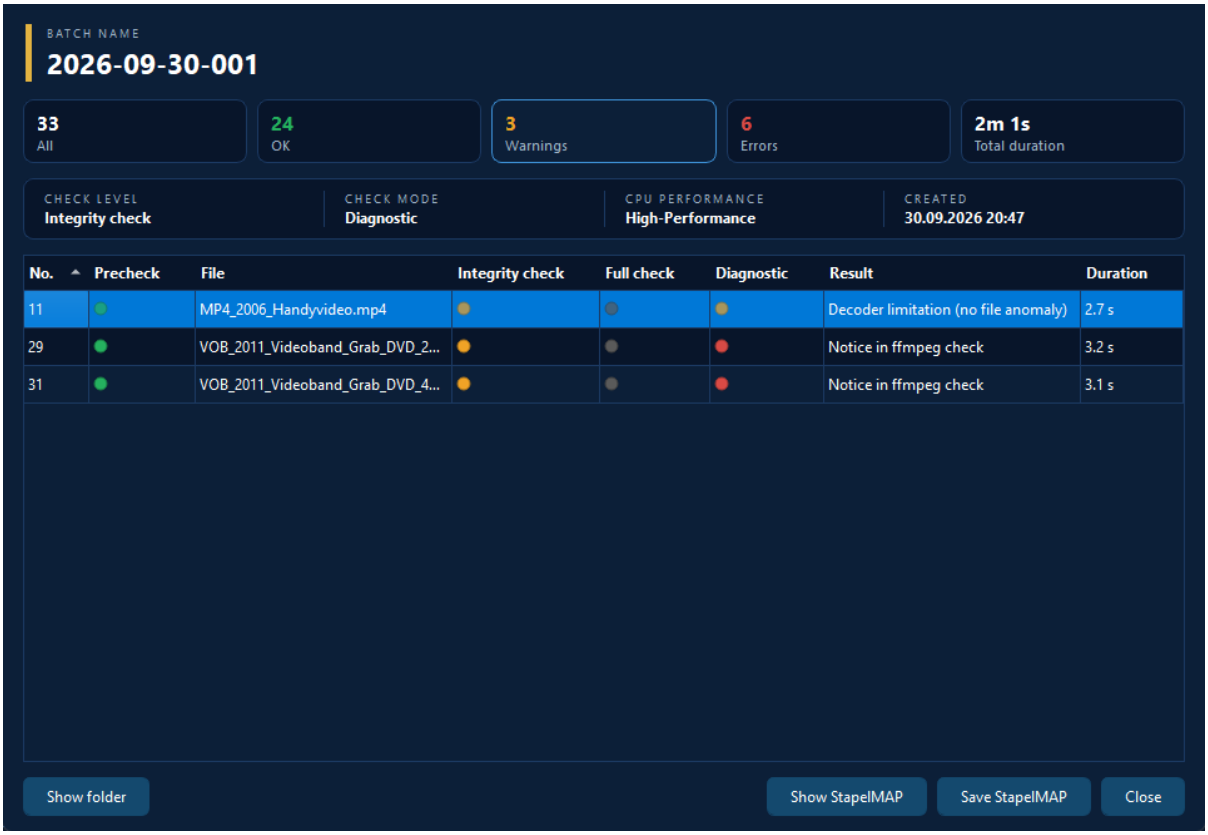


Figure 21: *The batch result with an active filter*

“Show StapelMAP” and “Save StapelMAP” generate the batch's check protocol as a PDF: one row per file with number, the results of the individual check levels, the finding in up to two lines and the duration, plus a legend, the active filter and a record box with rule set and tool versions. This makes the StapelMAP suitable as proof for acceptance, hand-over or archive audits (Chapter 10).

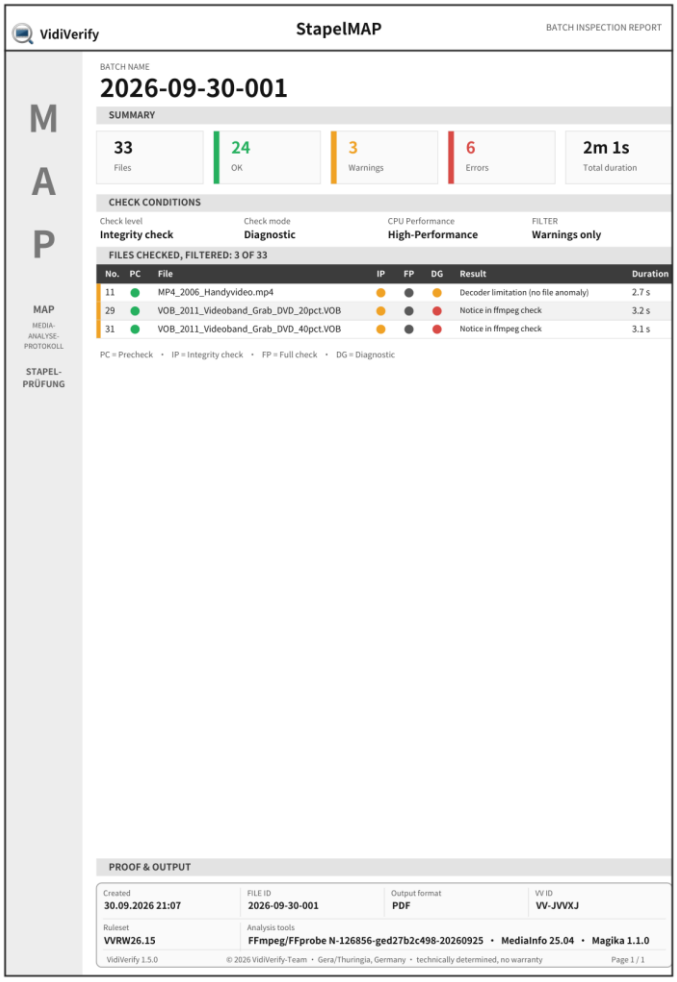


Figure 22: *The StapelMAP as a PDF*

7.6 Media Batch in FREE

FREE shows the complete workflow: the precheck runs for the first three files, the first file is checked in full, and the result window and StapelMAP are available. A notice then states the number of remaining files and offers “Unlock PRO”. Once PRO is unlocked, the run continues without a restart; VidiVerify catches up on the missing prechecks automatically.

8. Media Repair

A damaged file is often not lost. Media Repair picks up where the check ends: the finding becomes a well-founded repair path, carried out on a copy and then confirmed by a complete check. The original always remains untouched. Every repair is documented, from the command executed to the checksum of the new file.

> **PRO** **PRO feature**
>
> Media Repair is part of PRO. Once PRO is unlocked, VidiVerify automatically downloads the required MKVToolNix plug-in.

8.1 From finding to recommendation

Media Repair does not scan on its own; it reads the finding of the central check. After a mere precheck, the tab therefore asks: “Please check the file first”. The deeper the check, the more precise the recommendation; a run in the Diagnostic check mode works best. If there are no anomalies, the tab reports “No repair needed”.

The finding card shows the file name, classification, traffic light and a media overview with container, video, audio, bitrate, duration and size. Below it, VidiVerify rates every eligible method with a bar and a traffic-light dot and arrives at a well-founded recommendation. Methods are ranked by the severity of the finding, then by how many of the detected error types a method covers, and by its likelihood of success; the ranking is defined in the rule set. Methods that are unsuitable for this file remain visible, struck through. This makes it clear why a path is ruled out.

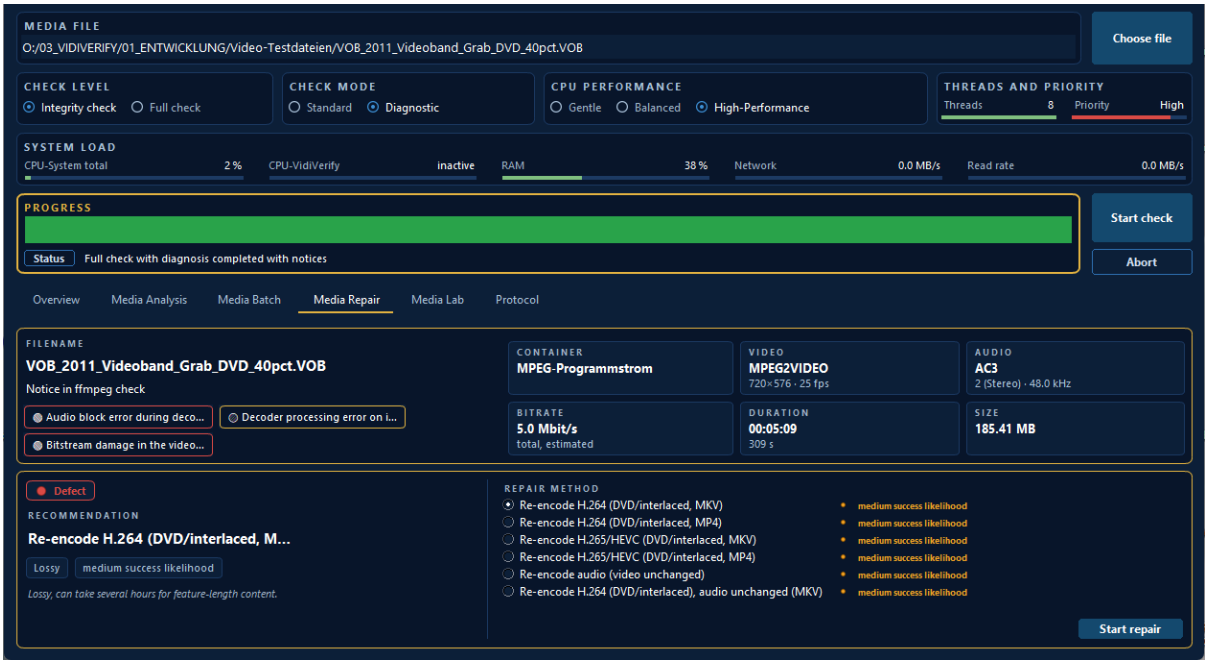


Figure 23: *Finding card and method selection with recommendation*

Media Repair can also be opened directly from the diagnostic report: with PRO, the button there reads “Start Media Repair”.

8.2 The repair methods

Media Repair offers 16 methods. Lossless paths come first; re-encoding is the last resort when the picture data itself is destroyed. Each method checks for itself whether it fits the file: the H.264 filter applies only to H.264, and audio re-encoding only if a short trial run on the audio track actually yields a signal.

Method	Type	Helps with
Re-mux with faststart (moov fix)	lossless	Repackages the file and moves the index to the start; typical after aborted recordings
Best-effort copy (skip errors)	lossless	Skips damaged sections and keeps everything readable
Regenerate timestamps	lossless	Jumps and gaps in the timeline
Bitstream filter (H.264 Annex-B)	lossless	H.264 data stream in the wrong format
Remux MKV (mkvmerge)	lossless	Damaged MKV structure, rebuilt with MKVToolNix
Re-encode H.264 or H.265, each normal or DVD/interlaced, as MKV or MP4 (eight variants)	lossy	Destroyed picture data that can be rescued only by re-encoding
Re-encode audio (video unchanged)	audio only	Defective audio track with an intact picture
Re-encode H.264, audio unchanged (also DVD/interlaced, MKV)	video only	Defective picture; all audio tracks are preserved

> **[!] Allow time**
> Re-encoding is compute-intensive and can take several hours for a feature-length film. The CPU performance setting from the main window applies here too; “Gentle” keeps the computer noticeably responsive.

8.3 The process in one card

“Start repair” opens a card that stays in place until the result. A stepper shows the phases Prepare, Repair, Verify and Done, with progress bar, method and elapsed time below. The live protocol next to it shows every step, including VidiVerify's own checks: free disk space, checksum of the input file, tool version used.

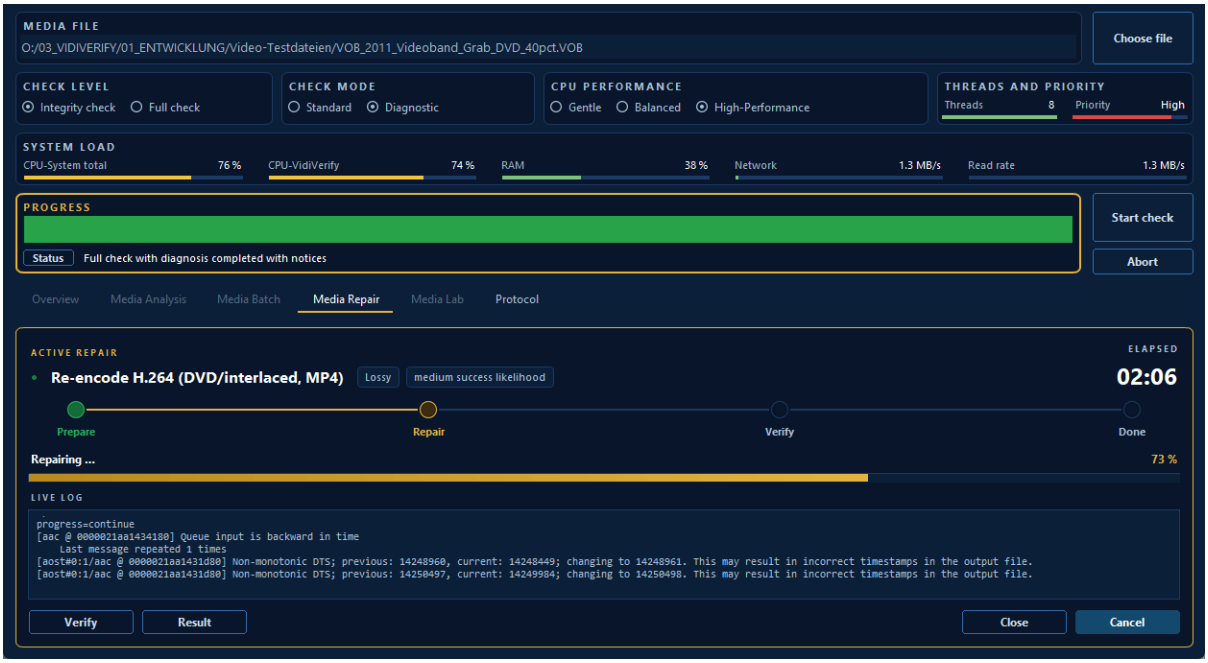


Figure 24: *A running repair with live protocol*

While the tool is writing, the other tabs (except Protocol), the CPU controls and the settings are locked. “Cancel” is available at any time. The repaired file is created next to the original and carries the method and timestamp in its name, for example “Urlaub.repair-remux_faststart-20260927-141233.mp4”.

8.4 Verification: mandatory, not optional

A tool that finishes without an error message does not yet prove a successful repair. That is why Media Repair stops after the tool run and reports “Repair completed”; the only way forward is “Verify now”. The repaired file then runs through the regular check at the selected check level. The verdict compares the finding before, taken from the deepest check already run, with the finding after:

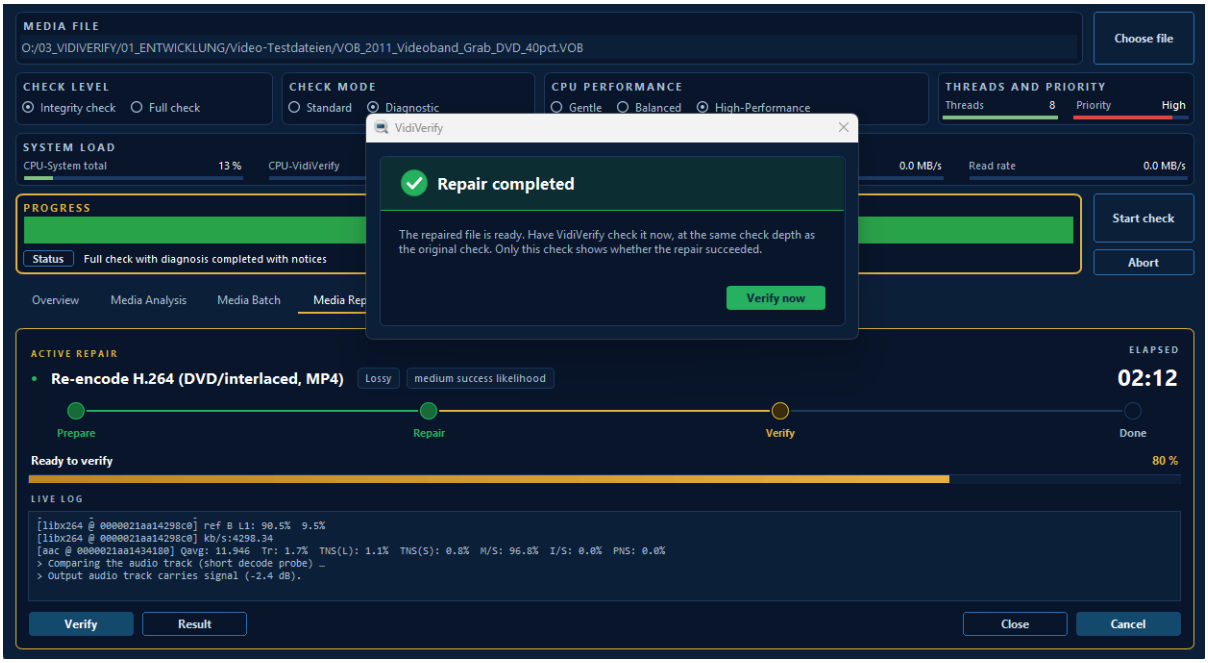


Figure 25: *After the tool run: “Repair completed”, continue only via “Verify now”*

Verdict	Meaning	The new file
Repair successful	The defect has been fixed	is kept
Partially improved	Better, but not completely, or the audio track has changed	VidiVerify asks: “Keep” or “Delete”
Strategy had no effect	The finding is unchanged	is removed
Repair introduced a new defect	The repair caused damage	is removed
Repair canceled	No verdict possible	is left in place

The check reads audio only as structure. A formally valid but silent audio track would slip past it. Media Repair therefore measures the level and samples of the audio track before and after the repair. If the track is missing afterwards, is silent, or there are fewer tracks than before, the verdict is “Partially improved”, with a note in the protocol and the result. Only what has been measured is stated: if the original was already silent, this statement is omitted.

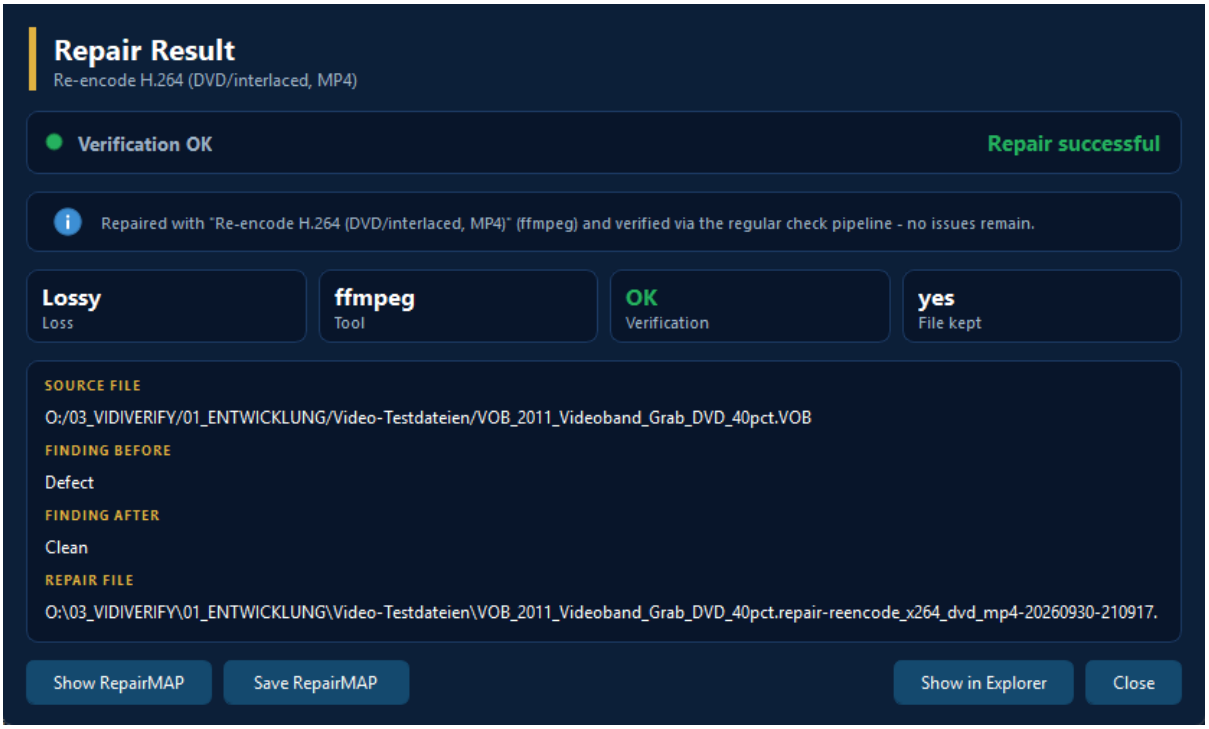


Figure 26: *The result after verification*

If a method was not enough, “Try another method” leads back to the selection.

8.5 RepairMAP: the audit sheet

The result window produces the RepairMAP, a single-page A4 dossier in the MAP layout. It records: the verdict with explanation, time frame and file size before and after, both files with full path, the method with tool, version and path, the finding before with up to three concrete anomalies and the finding after, the check conditions, the command line actually executed and the SHA-256 checksums of the source and target files. Anyone passing on a repaired file can thus prove what was done to it.

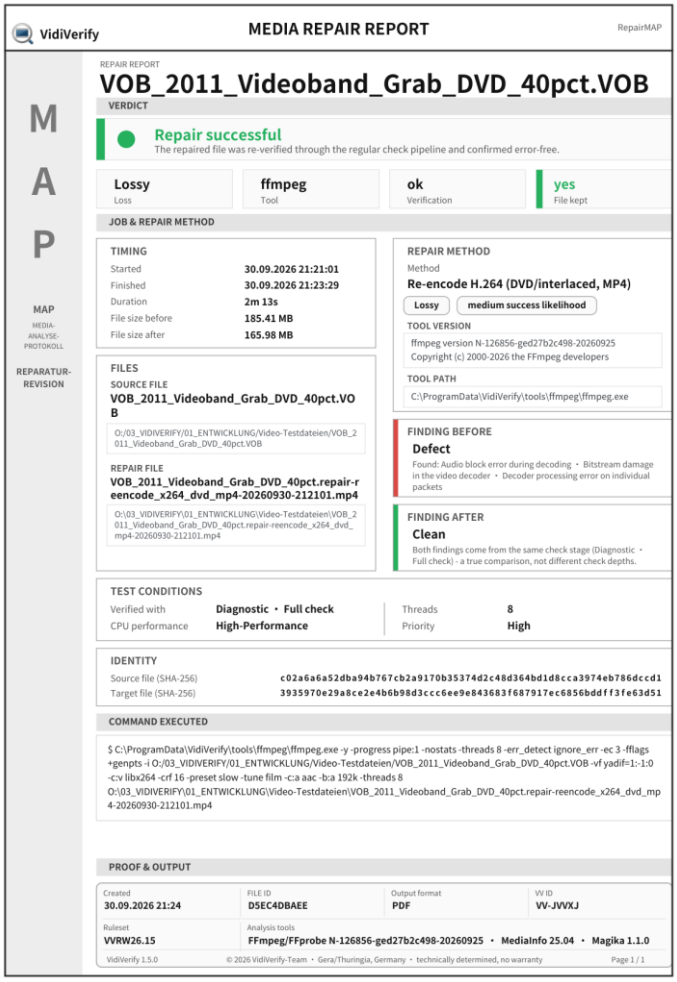


Figure 27: *The RepairMAP as a audit sheet*

8.6 Rescue, not invention

Media Repair recovers what a damaged file contains and stays true to the original: every rescued frame and every rescued sound comes from the file itself. Each method responds specifically to a concrete finding. If no valid container format can be detected at all, as with a renamed text file, VidiVerify recognizes this immediately and recommends checking the file or obtaining it again, instead of spending time on a hopeless repair.

> **PRO** **In preparation**
>
> Further repair methods, for example for correcting audio/video offset, arrive via update and are included in PRO.

9. Media Lab

Media Lab is a dedicated tab in the main window and the launchpad for VidiVerify's advanced, file-related tools, fully included in FREE. The tab is theme-aware and switches live with the selected interface design (Design mode, Windows classic, PLEX mode). It is divided into two areas: the compatibility analysis of the loaded file, and the tools and advisors.

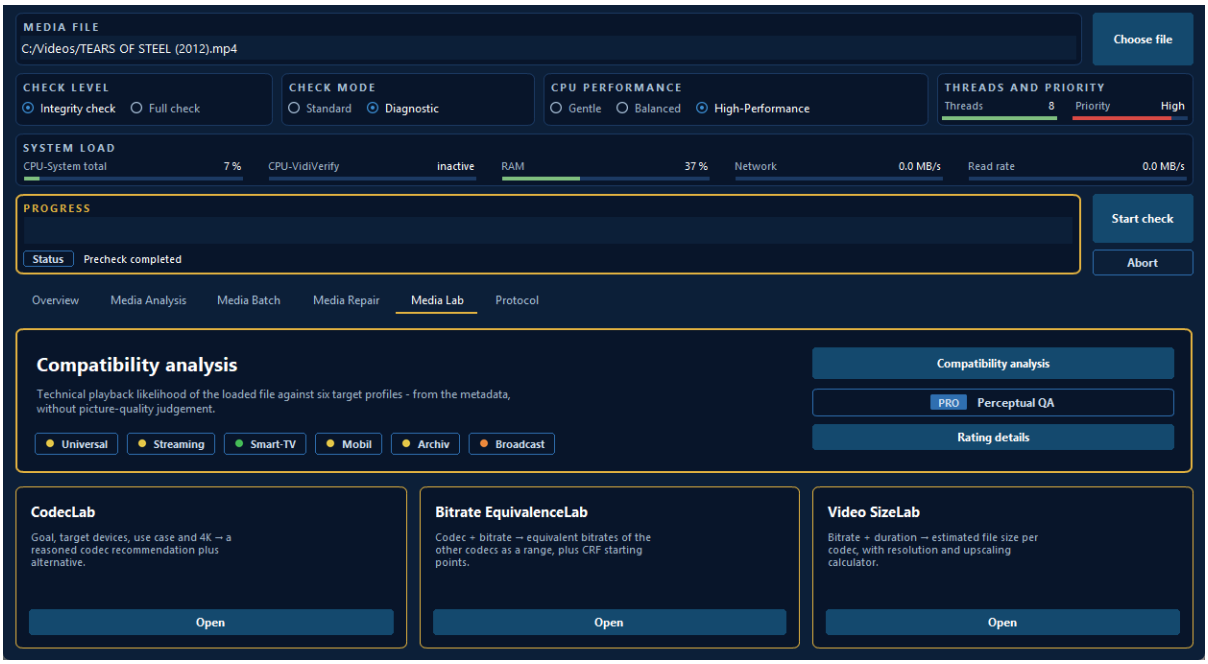


Figure 18: *The Media Lab tab: compatibility analysis, tools and advisors*

9.1 Compatibility analysis

The compatibility analysis rates a file's technical likelihood of playback against six typical target profiles, based solely on the harmonized metadata from FFprobe and MediaInfo. It answers the question that comes before every hand-over: will this file play where it needs to? Instead of a blanket “plays everywhere”, VidiVerify derives a technically grounded assessment for each target from objective file characteristics.

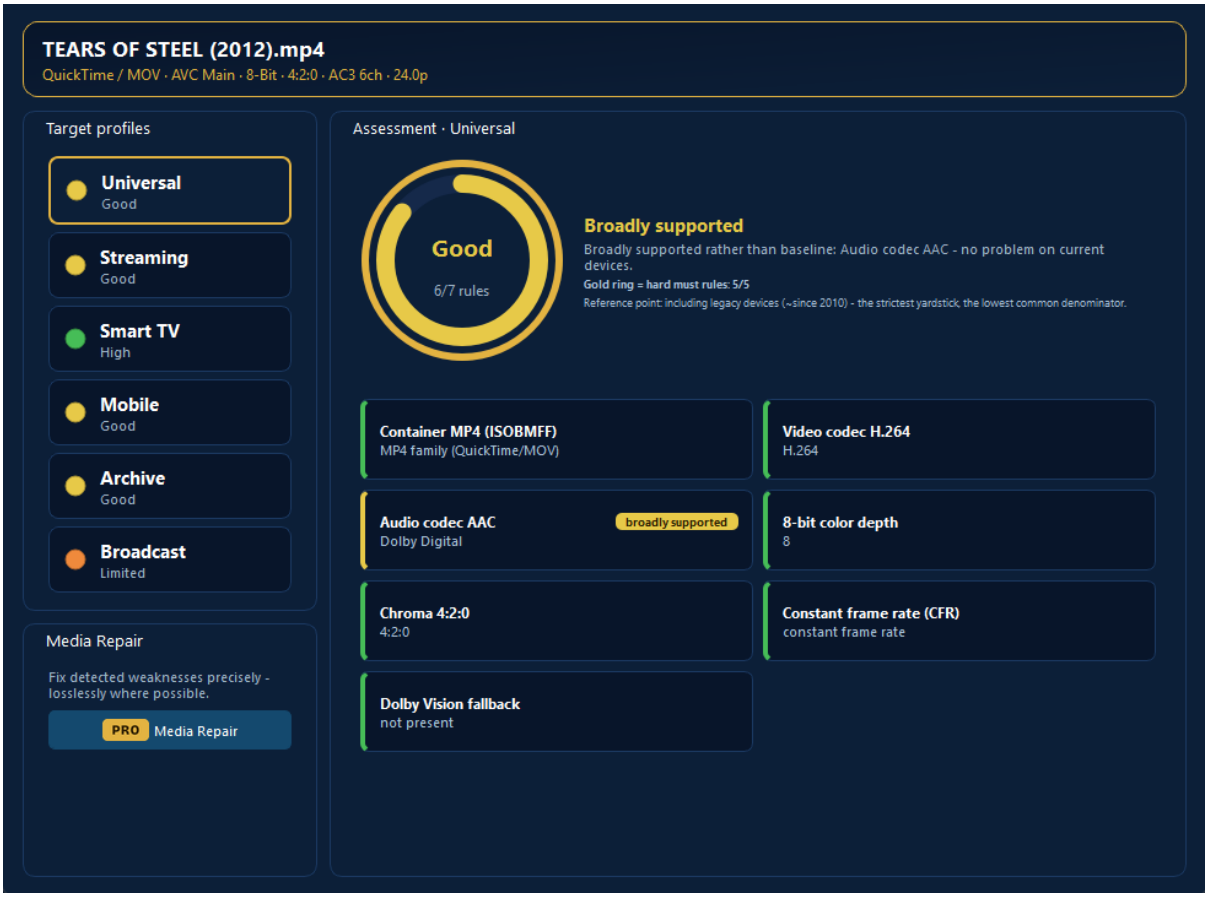


Figure 19: *The compatibility analysis: six target profiles, each rule with its requirement and the value found*

Target profile	Reference point
Universal	Broad playability on as many devices and players as possible
Streaming	Delivery via the web and streaming services (e.g. fMP4 with index)
Smart TV	Playback on standard smart TVs and set-top boxes
Mobile	Playback on smartphones and tablets
Archive	Long-term, lossless preservation
Broadcast	Professional broadcast and production environments

Each profile receives a traffic-light rating on four levels:

- Green (High): the file reliably meets the profile's requirements
- Yellow (Good, broadly supported): plays well, with minor limitations as a reference point
- Orange (Limited): playback likely only under certain conditions
- Red (Low): technically unsuitable for this profile

> **[i] Conservative, not sugarcoated**
> When characteristics are contradictory or unclear, the rating errs on the side of caution. Special cases are detected and named: Dolby Vision profile 5 (without an HDR10 base layer) shifts colors on players without Dolby Vision support and is therefore disqualified for Universal, Streaming and Mobile; fragmented MP4 without an index is a hard exclusion rule for Streaming. The assessment stays reliable, neither sugarcoating an intact file nor overlooking a problematic one.

The result appears in its own window with a profile bar and a double ring: the bold inner ring summarizes all rules in the respective traffic-light color, and the thin outer ring represents the hard must-have rules. Explanation tiles compare each characteristic's requirement with the value actually found, translated into plain language. The compatibility card in Media Lab also shows a short traffic light for the loaded file and becomes active as soon as a valid, prechecked file is loaded. A direct path leads from the window to the rule set (Section 10.6).

9.2 Tools and advisors

The second area brings together three compact tools that answer technical questions about codec, bitrate and resolution without requiring deep expertise.

- CodecLab: classifies codecs and their properties and helps choose the right format for a given purpose.
- Bitrate EquivalenceLab: translates bitrates between codecs: which bitrate in a more modern codec corresponds to a given bitrate in an older one, at comparable quality?
- Video SizeLab: estimates the effect of scaling. It distinguishes simple upscaling (UPS) from AI-based methods (VSR) and guides you from source to target format, letting you try out the bitrate with a slider.

> **PRO Media Lab keeps growing**
> Perceptual quality analysis for picture and sound will find its place here. It is in preparation and arrives via update, included in PRO.

10. The proprietary MAP (Media Analysis Protocol)

10.1 Concept and objective

The MAP (Media Analysis Protocol) is VidiVerify's proprietary centerpiece. It translates the results of deep technical check runs into a clear presentation any user can understand: color codes instead of columns of numbers, meaningful classification instead of raw values. At the same time, it is a record that documents which rules and which tools were used for the check.

> **[*] Technology meets clarity**
> Raw data from FFprobe, FFmpeg and MediaInfo often comprises thousands of values. The MAP distills them into statements you can grasp at a glance, and makes professional media analysis usable for everyone.

10.2 What the MAP delivers

- A summary of all check levels in one overview
- Color-coded presentation of the result (see Appendix B)
- Key technical values, clearly classified
- Anomalies in plain language
- A record of the rule set and tool versions
- Export as a designed, single-page A4 PDF archive sheet

10.3 Structure of the MAP

The MAP is a single-page A4 archive sheet. It is divided into clearly separated areas that together give a complete picture of the checked file:

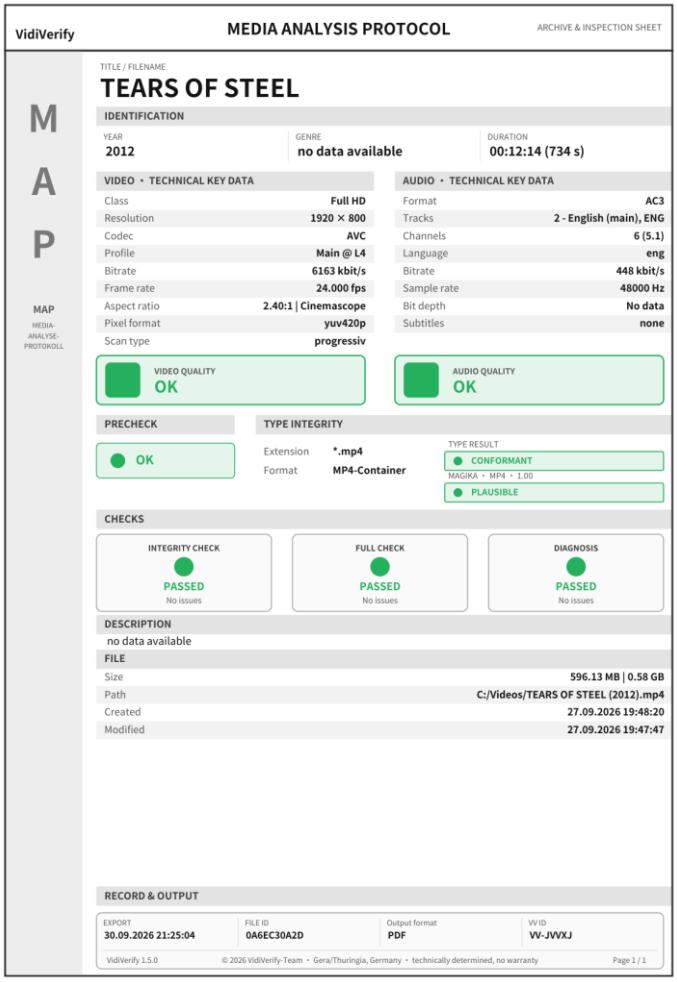


Figure 12: *A MAP as a PDF archive sheet*

Area	Content
Header	Logo, title “MEDIA ANALYSIS PROTOCOL” and file title
Identification	Year, genre and duration at a glance
Video and audio	Key picture and sound values, each with a colored verdict in the level color
Precheck and type integrity	Type result and Magika finding
Checks	Integrity check, full check and diagnostic with status, result and short text
Description	Classification and anomalies in plain language
File	Size, path, created and modified date
Record & output	Rule set ID and versions of the check tools; footer with creation date, file ID, output format and VV-ID

[*] Multilingual down to the font

The MAP embeds a Unicode font and renders German, Chinese, Cyrillic and Greek cleanly, without requiring a matching system font. The PDF still stays small.

10.4 MAP export

Save the MAP as a PDF via the “MAP” card in the “Media Analysis” tab. Preview and export use the same renderer, so the saved document matches the view exactly. The PDF is well suited for archiving, sharing and documenting checks.

10.5 The MAP family

PRO adds two more protocols. All three share the header, structure and record box, so they are recognizable at a glance as belonging together:

Protocol	Purpose	Described in
MAP	Check protocol for a single file	this chapter
StapelMAP (PRO)	Check protocol for an entire batch, one row per file	Section 7.5
RepairMAP (PRO)	Audit sheet for a repair, with command line and SHA-256 checksums	Section 8.5

PRO In preparation: MAP+

MAP+ will be the extended expert report of the MAP: multi-page, with content hash and chain of evidence, for archives, expert reports and client hand-overs. The card is already visible in the “Media Analysis” tab and arrives via update, included in PRO.

10.6 The rule set: the core of VidiVerify

Tools like FFmpeg or MediaInfo measure. What a measurement means is decided by the rule set. At what bitrate is a video adequate for its resolution? How can you tell that a FLAC file actually comes from an MP3? Which decoder message is a real defect, and which is merely an encoder quirk? The answers to such questions are VidiVerify's real knowledge. They are not contained in any plug-in or any language model; they grew out of innovation, experience and technical expertise.

What the rule set covers

- Bitrate rules for video and audio, per codec and resolution, with dedicated thresholds for formats such as HE-AAC
- Type integrity and the decision matrix for the AI plug-in Magika
- Container signatures and names, resolution classes and aspect ratio

- Edge measurement criteria for inflated bitrates and fake lossless files, plus the criteria for audio dropouts
- Classification of diagnostic findings by rank and traffic light
- Compatibility rules for six target profiles, each with condition and severity
- The traffic-light colors themselves: one color family for the entire application

A rule set that evolves

The rule set grows from real cases. Some converters write an MP3 tag in front of the audio data of a FLAC file; since then, VidiVerify recognizes FLAC and AAC even behind such tags, verified on 730 FLAC files from a CD collection. Listening tests and manufacturer specifications showed that HE-AAC sounds significantly better than AAC-LC at the same bitrate; since then, dedicated thresholds apply. A real DVD video showed that the same picture damage appears under different decoder messages; since then, VidiVerify classifies by the type of damage, not by the wording. With loudly mastered music, a simple cutoff frequency measurement is misleading; since then, VidiVerify looks for the edge an encoder leaves behind, validated on 731 CD rips. Each version is more precise than the one before.

Each version carries a fixed ID, for example “VVRW26.13”, and that ID appears in every protocol. When a rule changes, the ID changes; a MAP can therefore be classified unambiguously even years later. When the rule set was migrated to its current form, VidiVerify proved that not a single rating changed in the process: across more than 2.5 million combinations of codec and bitrate, 1,120 compatibility comparisons and 42 real test files.

Open for inspection

The “Rating details & symbols” window lays the rule set open, with a search bar: check basis with tool versions, levels and bitrate rules, special cases, type integrity, aspect ratio, resolution classes, container signatures, diagnostic findings and, for each target profile, the compatibility rules. With a file loaded, a value band shows where its bitrate falls between the thresholds. The window is available from the “Rating details” card in the “Media Analysis” tab and directly from the compatibility window.

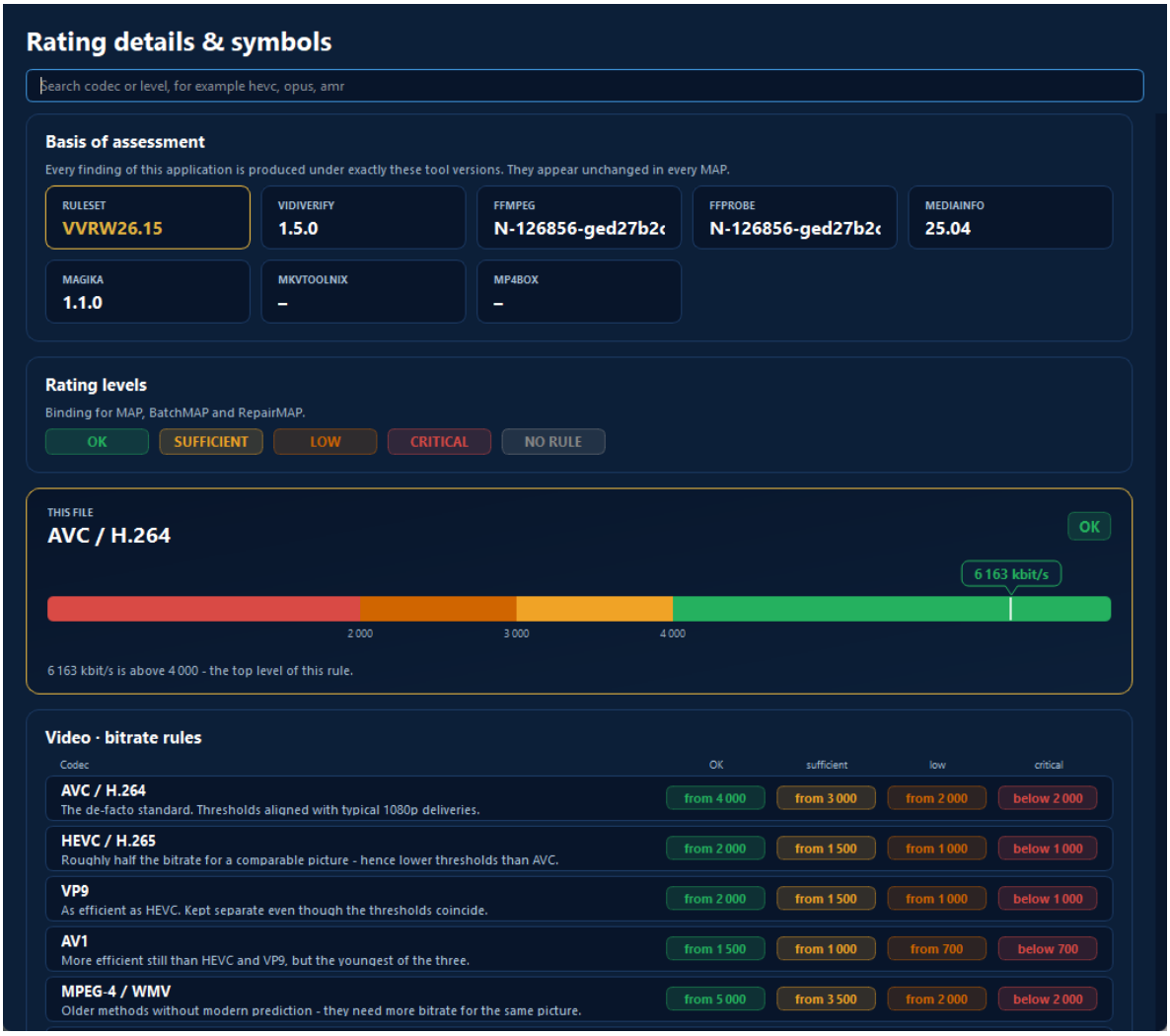


Figure 13: *The "Rating details & symbols" window*

[*] Whoever rates, discloses the rules

Many check programs deliver a verdict without a rationale, and an AI justifies itself differently every time. VidiVerify shows which rule it applied, documents the derivation and records the version. This curated, evolved knowledge is exactly what sets VidiVerify apart from a pure AI application (Section 1.6).

11. The integrated plug-ins

VidiVerify relies on selected, specialized plug-ins, each handling a clearly defined task. The application coordinates this ensemble, configures the tools automatically and brings their results together in the MAP. You work with a single, consistent interface, while proven tools run in the background whose combined insight goes far beyond what any single one could deliver.

> **[i] Only verified tools**
> VidiVerify checks exclusively with plug-ins it has obtained itself and verified against a fixed SHA-256 value. Every program version knows the tool versions it was tested with; if a plug-in is older, VidiVerify downloads it in full before the next check. Every protocol states the versions actually used.

11.1 FFmpeg

FFmpeg is the globally recognized tool for processing and decoding media files. In VidiVerify, FFmpeg performs the actual decoding in the integrity check and full check, as well as most Media Repair methods. VidiVerify uses a GPL build that also includes the encoders for H.264 and H.265.

- Core task: decoding and checking all streams, repair
- Strengths: broad format support, very robust error detection
- Used for: integrity check, full check, diagnostic, Media Repair

11.2 FFprobe

FFprobe is part of the FFmpeg package and reads metadata and stream information. VidiVerify uses FFprobe in the precheck and for the basic technical data in the MAP.

- Core task: reading containers, streams and codec information
- Strengths: fast analysis without full decoding
- Used for: precheck, diagnostic, metadata in the MAP

11.3 MediaInfo

MediaInfo provides particularly detailed information on container, codecs and advanced parameters. VidiVerify uses MediaInfo as a second source, for cross-checking and as the basis of the compatibility analysis. MediaInfo also detects Dolby Atmos and DTS:X structures. With multiple audio tracks, main-track selection takes language and default flag into account: a German interface finds the German audio track.

- Core task: in-depth metadata analysis
- Strengths: very detailed technical information, Atmos and DTS:X detection
- Used for: cross-checking, compatibility analysis

11.4 Magika

Magika is an AI-powered tool for content-based file type detection, released by Google and used by Google itself in its security pipelines. While the classic magic-byte check reads only the file header, Magika evaluates the content. VidiVerify places both results side by side: what one method misses, the other catches. For disguised files, such as a PDF with the extension .mp4, VidiVerify clearly marks the result as “non-media” with a note on how to correct it.

- Core task: content-based file type detection
- Strengths: robust even with renamed or manipulated files
- Used for: type integrity, second opinion on the magic-byte check

11.5 MKVToolNix

MKVToolNix is the standard tool for the Matroska container. VidiVerify uses its mkvmerge component to rebuild damaged MKV structures losslessly. The plug-in belongs to Media Repair; once PRO is unlocked, VidiVerify downloads it automatically from its own verified mirror.

- Core task: rebuilding MKV containers
- Used for: Media Repair (PRO), method “Remux MKV (mkvmerge)”

11.6 VidiVerify Runtime (PyAV and OpenCV)

The graphically demanding windows, from the detail windows and Media Lab to the result windows of Media Batch and Media Repair, run in their own runtime environment: the Qt-based VidiVerify Runtime. It is a bundled package that is downloaded automatically when first needed, with named progress and cancelable at any time. Each program version receives its own matching runtime.

The runtime contains high-quality media libraries: PyAV binds the complete FFmpeg library for frame-accurate decoding, and OpenCV, the industry standard for computer vision, provides the foundation for image and frame analysis and, in the future, for perceptual quality analysis.

- Core task: detail, tool and result windows as well as image and frame analysis
- Contains: PyAV, OpenCV and the Qt interface

11.7 Download and updates

VidiVerify downloads all plug-ins itself, on request already at first launch. Every plug-in has a second download source; if the first is unreachable or noticeably slow, VidiVerify switches automatically. Downloads can be canceled at any time. An update is built and verified alongside the previous version; if it fails, VidiVerify restores the previous version. The installation stays fully functional even if an update is interrupted.

12. Configuration and parameters

Open the settings via “Menu > Settings”. A navigation rail on the left divides them into five sections: Check behavior, Display & Design, Plug-ins, Updates, and Storage & Maintenance. VidiVerify saves all changes in the file vidiverify_settings.json in the user folder. Settings are locked while a repair is running.

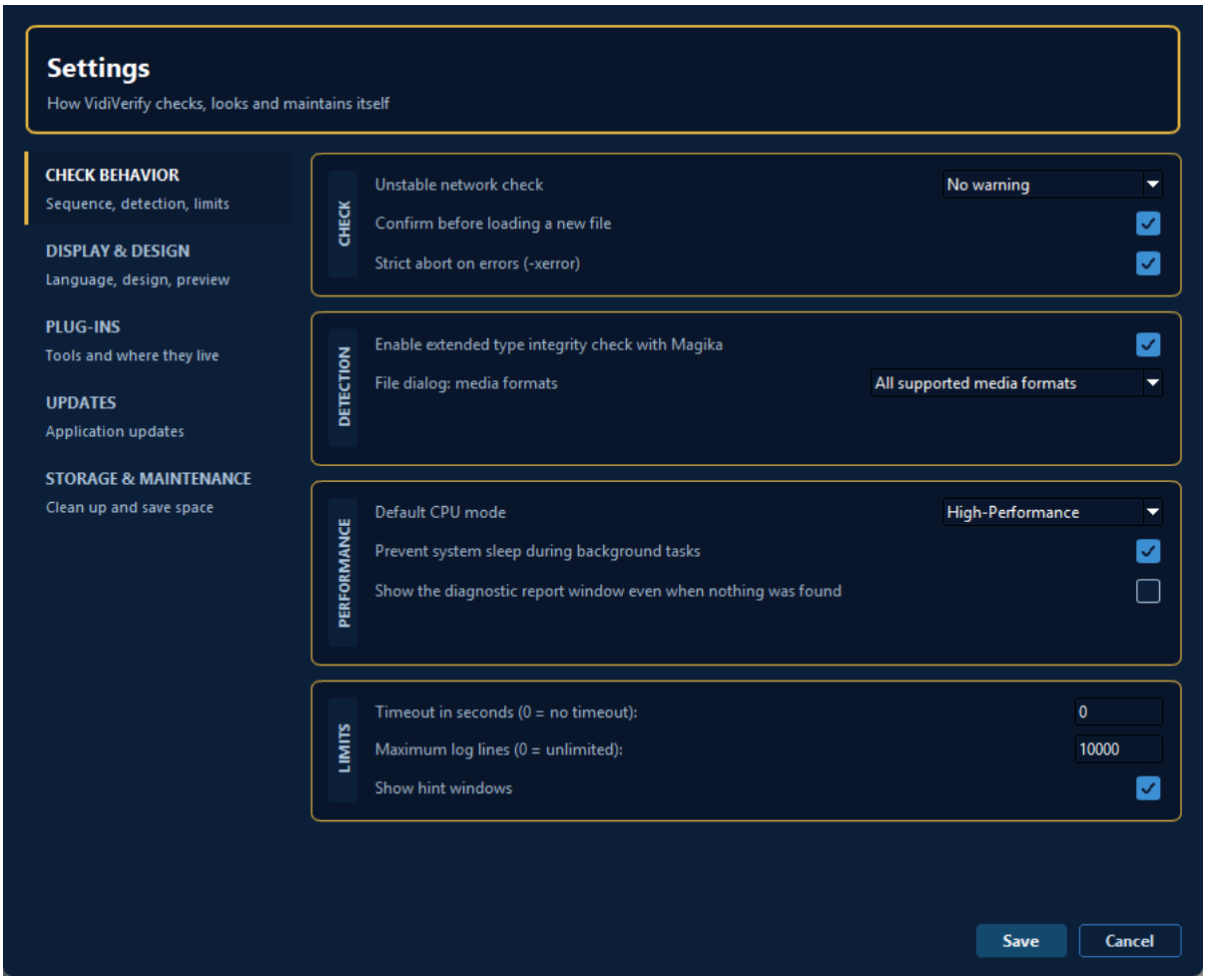


Figure 14: *The settings with navigation rail*

12.1 Check behavior

Setting	Description	Default
Strict abort on errors	The first decoding error ends the check	On
Enable extended type integrity check with Magika	A second, content-based opinion on the file type	On
Default CPU mode	Performance mode at program start (Section 12.2)	Balanced

Setting	Description	Default
Show the diagnostic report window even when nothing was found	Opens the diagnostic report even for an unremarkable file	Off
Show hint windows	Explanatory hints, for example when an audio file is loaded for the first time	On
Timeout in seconds	Time limit per check; 0 means no limit	0
Maximum log lines	Upper limit for the runtime protocol; 0 means unlimited	10,000
Unstable network check	Handling of files on unreliable network paths: No warning, Warn only, Block check	Warn only
File dialog: media formats	Video formats only, audio formats only, or all	All
Confirm before loading a new file	Protects an existing result from being replaced by accident	On
Prevent system sleep during background tasks	Keeps the computer awake during checks, batches, repairs and downloads; the display may turn off	On

12.2 Performance

CPU usage follows three performance modes that balance check speed against system responsiveness. You can switch the mode in the main window; it also applies to Media Batch and Media Repair.

Performance mode	Processor cores	Priority	Typical use
Gentle	one core	Low	Check or repair in the background during normal work
Balanced	half of the cores	Normal	Default for most cases
High-Performance	all cores	High	Fastest possible check, for example overnight

12.3 Display & Design

Setting	Options
Language	Deutsch, English, 中文, Español, Русский
Design	Design mode, Windows classic, PLEX mode
Compact mode	Automatic, On, Off: space-saving layout for small notebook screens (around 1280 × 800), takes effect after a restart
Scrollbars	Automatic (shown while scrolling), Always show, Never show

Three designs are available, each fully styled, from the main window to every dialog:

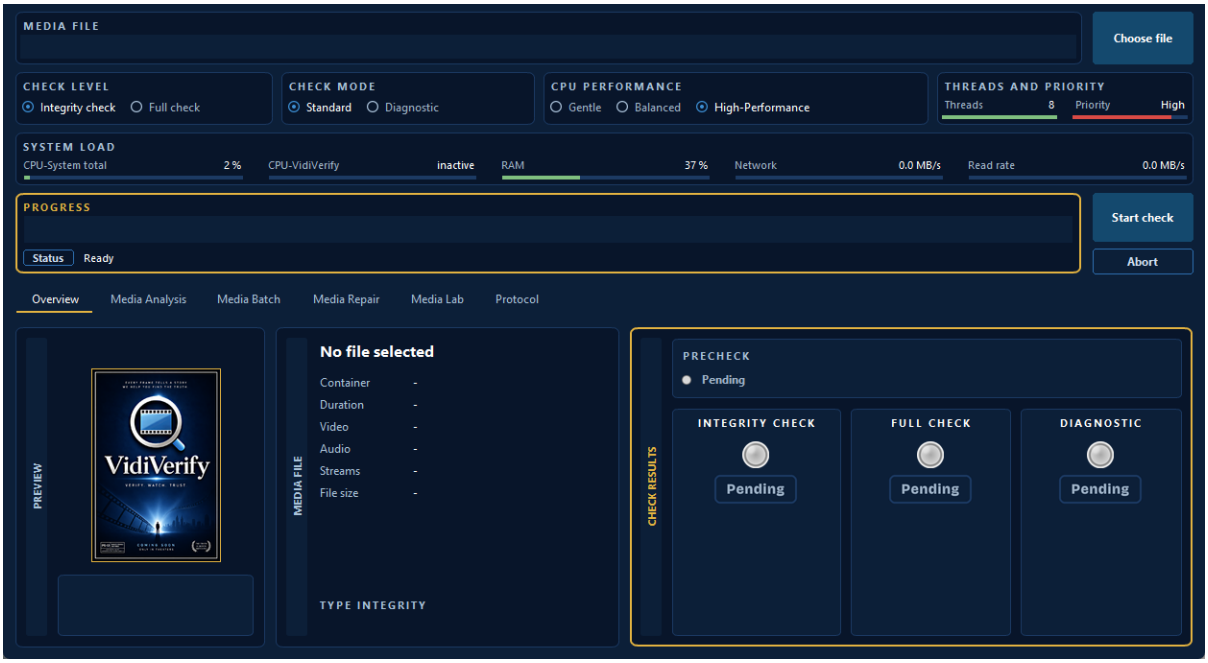


Figure 15: *Design mode: deep blue with gold accents*

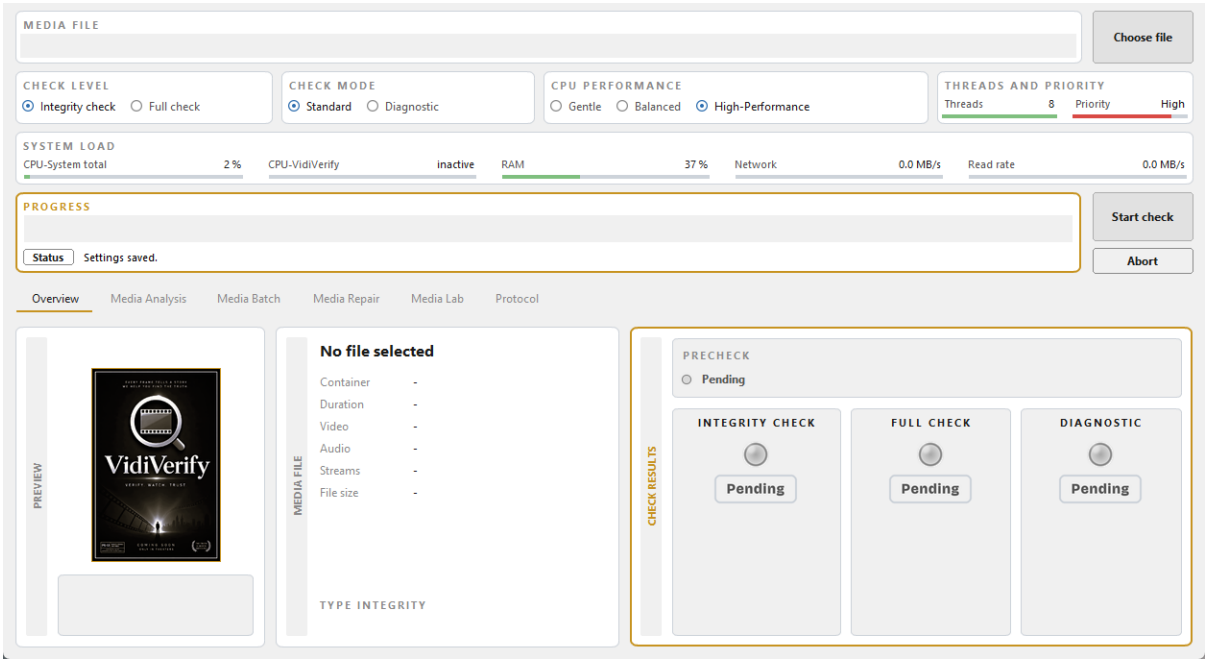


Figure 16: *Windows classic: light and clean, in the style of Windows*

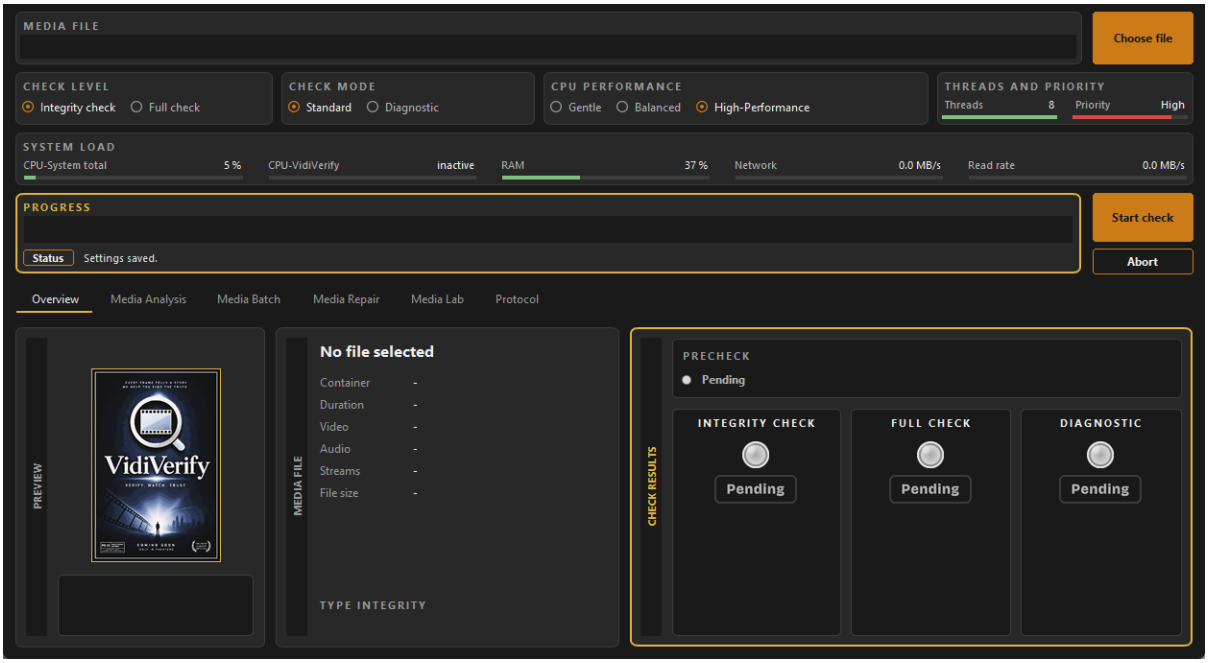


Figure 17: *PLEX mode: dark anthracite with orange, inspired by media servers*

12.4 Plug-ins

The “Plug-ins” section shows the version, status and actual path of every plug-in. Paths are managed by VidiVerify and cannot be pointed to other tools, so every protocol reliably states which tools performed the check. From here you can also download or update plug-ins individually.

12.5 Updates

- Check for VidiVerify updates at startup (can be turned off)
- Check for plug-in updates at startup (can be turned off), plus the “Check plug-ins now” button
- Release notes as an overview: each version as a card, the installed one highlighted; without a connection, the most recently loaded state remains visible
- Before updating to a new major version, VidiVerify asks for confirmation, because a PRO license does not cover the next major version (Section 3.2)

12.6 Storage & Maintenance

Here VidiVerify cleans up folders that grow over time, automatically at program start and at the push of a button, with adjustable scope:

- FFmpeg update backups: number of backups kept (default: the two most recent)
- Runtime packages: older versions by last use (default: keep two); the current one is always kept
- Preview image cache: size limit in megabytes (default: 200 MB)
- Diagnostic dumps (hang dumps): number of files kept (default: 20)

- Temporary update files: deleted once they are older than five minutes

“Clean up now” triggers the cleanup at any time; VidiVerify first shows the expected space to be freed, then the result. Each area can be set to “keep all”.

12.7 Support package

At the push of a button, VidiVerify assembles a support package: program version, plug-in versions, runtime protocols and the anonymized device ID (VV-ID). It contains no media content; on request, VidiVerify masks paths, computer names and network addresses. The package is assembled locally, and you decide whether to send it.

> **[+] Safe defaults**
> VidiVerify ships with defaults that suit most use cases. Changes are optional and can be reverted at any time.

13. File structure and storage layout

VidiVerify keeps three areas cleanly separated: the program installation, each user's data, and the plug-ins shared by all users of the computer. Settings, protocols, the preview cache and saved batches live in the personal user folder (%LocalAppData%). The plug-ins and the VidiVerify Runtime live machine-wide under %ProgramData%. This keeps the installation well organized and lets updates and uninstalls run cleanly.

13.1 General description

Program folder

```
\VidiVerify
├── VidiVerify  (main program)
├── VidiVerify-QuickCheck  (Explorer quick check)
└── VidiVerify Uninstall
```

User folder (per user)

```
\VidiVerify
├── Settings
├── Protocols and diagnostics
├── Preview cache and inbox
└── Saved batches
```

Plug-in folder (machine-wide)

```
\VidiVerify
└── tools (FFmpeg, MediaInfo, Magika, MKVToolNix, vv_runtime)
```

13.2 Example structure with file names

User folder (per user)

```
%LocalAppData%\VidiVerify
├── vidiverify_settings.json
├── module_sources.json
├── logs
│   ├── hangs
│   ├── exceptions.log
│   ├── faulthandler.log
│   ├── runtime.log
│   └── subprocesses.log
├── preview_cache
├── inbox
└── batches
    └── <batch>.vva
```

Plug-in folder (machine-wide)

```
%ProgramData%\VidiVerify
└── tools
    ├── ffmpeg
    ├── mediainfo
    ├── magika
    ├── mkvtoolnix
    └── vv_runtime
        └── <program version>
```

13.3 Notes on the key files

File or folder	Purpose
vidiverify_settings.json	Settings (language, design, performance mode, check behavior)
module_sources.json	Approval of the individual plug-ins
logs/runtime.log	Chronological events from live operation
logs/exceptions.log	Exceptions with full stack trace
logs/subprocesses.log	Log of the plug-in processes started
logs/faulthandler.log	Information on severe crashes
logs/hangs	Diagnostic dumps for unresponsive processes
preview_cache	Cached preview images
inbox	Handoff folder for files passed in from Explorer (QuickCheck and “Open with”)
batches	Interim states and saved batches from Media Batch (extension .vva)
tools\ffmpeg	FFmpeg and FFprobe
tools\mediainfo	MediaInfo
tools\magika	Magika for type integrity
tools\mkvtoolnix	MKVToolNix for Media Repair (PRO)
tools\vv_runtime	VidiVerify Runtime, one folder per program version

[*] Transparency as a principle

Settings and protocols are stored in open plain-text formats (JSON, text). That makes them traceable outside the application as well, and easy to evaluate or archive exactly as you need.

14. Architecture and technical details

14.1 Overview

VidiVerify is developed in Python and uses the bundled standard library along with a selection of established third-party libraries. The main interface is built on Tkinter, Python's standard UI toolkit. The actual media analysis calls established, specialized command-line plug-ins (FFmpeg, FFprobe, MediaInfo, Magika). The graphically demanding windows, from the detail windows and Media Lab to the result windows of Media Batch and Media Repair, run in their own proprietary runtime environment: the VidiVerify Runtime, built on Qt. It is loaded automatically the first time it is needed and comes with a selection of high-quality Python media libraries (including PyAV and OpenCV) ready bundled. This keeps the lean main application small, while the compute-intensive analysis and rendering tasks run in an environment optimized for them.

14.2 Component overview

Component	Role
User interface (Tkinter)	Input, display, operation and presentation of results
Check module	Orchestrates the precheck, integrity check and full check as well as the diagnostic
Rule set register	Single source of all rating rules, with a versioned ID
Batch control (PRO)	Runs Media Batch, saves interim states and the tool versions used
Repair agent (PRO)	Runs Media Repair as a complete process: prepare, repair, verify
MAP (Media Analysis Protocol)	Proprietary evaluation and color coding of the results
Plug-in manager	Manages download, update and launch of the plug-ins
VidiVerify Runtime (Qt)	Its own runtime environment, loaded on demand, for detail, tool and result windows
Media libraries (PyAV, OpenCV)	Professional tools bundled in the runtime for decoding as well as image and frame analysis
Subprocess manager	Launches and monitors the plug-in processes
Logging system	Writes several levels of protocols to files in a structured way
Settings store	Persists user settings in JSON format
Watchdog	Monitors the user interface for unresponsive states
License module	Verifies signed licenses and unlocks PRO
Support package module	Automatically assembles diagnostic information

> **[*] Professional tools under the hood**
> With the VidiVerify Runtime, VidiVerify ships a selection of professional media libraries right out of the box: PyAV integrates the complete FFmpeg library for frame-accurate decoding, and OpenCV, the industry standard for computer vision, provides the foundation for the image and frame analyses (sharpness, motion, artifacts, continuity). VidiVerify orchestrates these high-quality tools with precision and translates their output into clear, understandable results.

14.3 Plug-in dependencies

- FFmpeg: decoding and checking of media files
- FFprobe: reading metadata and stream information
- MediaInfo: in-depth media analysis and supplementary metadata
- Magika: content-based file type detection for type integrity
- MKVToolNix: rebuilding MKV containers for Media Repair (PRO)
- VidiVerify Runtime (Qt/PyQt-based): runtime environment for the advanced windows, bundling the following media libraries
- PyAV: Python binding to the FFmpeg libraries for frame-accurate decoding (in the runtime)
- OpenCV: image processing and frame analysis for the detail windows (in the runtime)
- PyQt and pyqtgraph: graphical rendering and interactive charts in the runtime windows
- psutil: detailed CPU and process information
- Pillow: image processing for the interface
- tkinterdnd2: drag-and-drop support
- fpdf: generation of the PDF protocols
- truststore: certificate validation through Windows for all downloads

14.4 Concurrency

The application uses several layers of concurrency to stay responsive. The user interface runs on the main thread, while checks run on separate worker threads. Threads communicate through a central message queue, which keeps updates to the interface consistent.

14.5 Error handling

Errors are handled and logged at four levels:

Level	Handling
Operational notice	Informational message in the status bar and the protocol window
Warning	Visibly highlighted in the protocol, the check can continue
Error	The check is aborted, a detailed protocol entry is written

Level	Handling
Exception	Full stack trace in exceptions.log, diagnostic dump if needed

14.6 Single-instance mechanism

The application prevents multiple instances from running at the same time, using a system-wide lock (mutex) plus a supporting lock file. If you start a second instance, a notice tells you that the application is already running. Stale lock files are cleaned up automatically if the original process no longer exists.

14.7 Stable device identifier (VV-ID)

For support purposes, the application generates a stable, anonymized device identifier: the VV-ID (VidiVerify ID). It begins with the prefix VV- followed by five characters from a character set without look-alike characters. The identifier is based on a system-wide hardware identification and a built-in salt value. It cannot be traced back to individuals or specific hardware.

14.8 Updates and integrity

The application and the plug-ins are obtained only from signed or hash-verified sources. VidiVerify checks every package against its SHA-256 value before installing it; an update without a checksum is never even downloaded. The program files are signed with a code-signing certificate.

- Certificate validation by Windows itself, including on freshly set-up computers and in corporate networks with their own certificate authority
- Before every plug-in launch, VidiVerify checks the path for hidden redirections, so no foreign program can be slipped in
- Updates to plug-ins and the runtime are built and verified alongside the previous version; if they fail, the previous version stays in place
- The self-update setup runs from its own, unpredictable directory and is removed afterward

> **[*] Openness and verifiability**
> All formats, storage paths and protocol structures are designed to remain readable and analyzable even without the application. You stay fully in control.

15. Examples and use cases

15.1 Use case: intake inspection in a media archive

A media archive receives new files every day from production teams and external suppliers. Before a file is admitted into the archive, it must be checked for structural integrity. Media Batch checks a whole day's intake overnight in a single run; by morning, the StapelMAP is ready as documented proof.

Typical setup: Media Batch (PRO), full check, High-Performance mode, strict abort on errors, StapelMAP for filing.

15.2 Use case: quick check before handoff

An editor wants to briefly check a finished episode before delivering it to the client. A complete review is not needed, but typical container errors must be ruled out. The integrity check delivers a reliable interim result within a few minutes.

Typical setup: Integrity check, Balanced mode, MAP PDF for internal filing.

15.3 Use case: spot checks in an archive audit

An archivist regularly checks samples from the collection to detect gradual media aging. Every checked file receives a MAP as a PDF, which is attached to its entry in the collection management system. Because every MAP carries the ID of the rule set, checks from different years remain comparable.

Typical setup: Full check, Gentle mode during working hours, MAP as an archive document.

15.4 Use case: forensic review

A professional review in a forensic context requires traceable checks with complete documentation. VidiVerify logs every check with runtime, timestamp, result and classification. The Protocol tab and the subprocess log are additionally available as a text-based chain of evidence.

Typical setup: Full check with the Diagnostic check mode, strict abort on errors, maximum log lines 0 (unlimited), MAP as an archive document.

15.5 Use case: rescuing a family video

A digitized recording from the nineties stops playing after a few minutes. The integrity check reports an error, and the diagnostic names the cause: damaged timestamps. Media Repair recommends regenerating the timestamps, losslessly. After verification, the verdict reads “Repair successful”, and the RepairMAP records what was done. The original stays untouched right next to it.

Typical setup: Integrity check with the Diagnostic check mode, followed by Media Repair (PRO) with a lossless method.

15.6 Use case: routine IT maintenance

An IT department uses VidiVerify to regularly check media holdings stored on file servers. The goal is to detect storage problems early, before entire generations of backups are affected. The update check for the plug-ins stays active, so current decoders are always available.

Typical setup: Balanced mode, update checks enabled, Media Batch (PRO) for checking entire directories in series.

16. Planned extensions and outlook

VidiVerify is under continuous development. The following projects are designed or in preparation, most of them as PRO modules that arrive by update and are included in PRO at no extra charge. Scope and order follow demand and development progress, which is why this manual names no dates.

16.1 Media Compare PRO

Media Compare checks two versions of the same video or audio file and puts them side by side: the duplicate in the archive, the FLAC against the MP3, the second download against the first. What gets rated is not the individual file but the difference, and VidiVerify recommends which version to keep.

- Comparison by category, both files side by side
- A recommendation in plain words: quality leader and best overall version
- Also available from Media Batch for selected files

16.2 Music Batch PRO

Music Batch checks entire music collections, such as DJ libraries with tens of thousands of tracks, for misreported bitrates, defects and sound flaws. It groups duplicate tracks together and recommends which file to keep, based on measured rather than stated quality. A “320” that sounds like 128 kbit/s then loses out to a genuine 256.

16.3 Perceptual quality analysis for picture and sound PRO

Perceptual quality analysis adds perception to the technical check: how good does the file actually look and sound, regardless of its specifications? For the picture, it uses the image analysis of the VidiVerify Runtime; for the sound, loudness according to EBU R128, spectral evaluation and the search for anomalies.

- Visual quality assessment from the viewer's perspective
- Audio assessment with proven thresholds instead of estimates

16.4 VVProgressiv™: distributed, adaptive checking PRO

VVProgressiv™ is a specialized method developed by VidiVerify: an intelligent sequential check with automatic escalation. Instead of checking only the beginning of a file or decoding it completely for hours, VVProgressiv™ distributes short check windows across the entire runtime. Wherever something looks suspicious, the check depth increases right there, step by step, until the finding is certain.

The result is a reliable statement about the whole file in a fraction of the time a full check takes. For long recordings, archive holdings and batches in particular, this changes the equation: a defect in the

second hour no longer slips through the net of the first 30 seconds. And because VVProgressiv™ reports every area it checked, you always know exactly what the verdict is based on.

- Check windows distributed across the entire runtime
- Automatic escalation at suspicious points, step by step
- Transparent reporting of the areas checked
- Check logic developed specifically for VidiVerify

16.5 MAP+ as an expert report PRO

MAP+ will be the detailed, multi-page version of the MAP: with content hash, chain of evidence and the results of all modules, for archives, expert reports and client handoffs.

16.6 Additional repair methods PRO

Media Repair gains further methods, such as correcting audio/video offset and fixing faulty signaling, also accessible directly from the video detail window.

16.7 AI forensics PRO

Is this video real? Online scanners answer with a percentage, and in our tests they got it wrong: a genuine industrial video was flagged as a deepfake, and a fully generated video was rated as real. VidiVerify's AI forensics takes a different path of its own. It treats technical signs of AI generation or editing as indicators and strictly separates the indication level (how many characteristics point to it) from the confidence level (how reliable that assessment is).

New measurement channels enter the assessment only once they have proven their discriminating power on real material: calibrate first, then evaluate. The result is an assessment you can rely on, because it never claims more than the file actually reveals. The analysis runs entirely locally; no file ever leaves the computer.

- Evidence-based assessment instead of a blanket percentage
- Indication level and confidence level always separate and always reported together
- Only calibrated measurement channels count
- Fully local

16.8 Statistics module PRO

The statistics module records real check data and learns from it how long checks take on your specific computer, depending on system, performance mode and storage location.

16.9 YTUP™: YouTube upload forecast PRO

Every video that lands on YouTube is re-encoded there, and what remains of the original only becomes visible when it is too late. YTUP™ reverses the order: before the upload, VidiVerify assesses the file technically and qualitatively and forecasts how it will look after the platform's re-encoding. This is not a format check, but a realistic prediction of the visible quality.

For content creators, agencies and newsrooms, this means choosing bitrate, resolution and codec so that the best possible result arrives after the upload, instead of wondering about mushy images afterward.

- Assessment of upload suitability
- Forecast of image quality after re-encoding
- Concrete recommendations for optimizing before the upload

16.10 DVD and Blu-ray checking

A dedicated check for video and data DVDs as well as Blu-ray structures, going beyond the individual file.

16.11 NFO export for media management

The check results will produce an .nfo file that media management systems such as Kodi, Jellyfin and Emby can read. Your media library gets technically accurate information with no upkeep.

16.12 VidiVerify as a tool for AI assistants

AI assistants will be able to call VidiVerify on their own through a documented interface. The AI asks the question and explains the result, and VidiVerify delivers the verified finding (see Section 1.6).

16.13 macOS and more languages

VidiVerify is set to arrive on macOS as well, with consistent operation and a comparable feature set. More interface languages are planned, including Hindi and Portuguese.

> **[*] Outlook**
> These projects make VidiVerify a comprehensive platform for checking, rating, repairing and managing media files. The modular architecture allows them to be introduced step by step, without compromising the stability and simplicity of the application.

Appendix A: Glossary and Terms

The following overview explains key technical terms from the fields of media files, codecs, audio properties and the integrated plug-ins, as far as they are relevant to VidiVerify. The terms are listed in alphabetical order.

Term	Explanation
AAC	Advanced Audio Coding, a lossy audio codec, widely used in MP4 and AAC streams
AC3	Audio codec from Dolby, common on DVD and in broadcasting
AI forensics	Planned PRO module that weighs technical indications of AI involvement in video files, separated into AI indication level and confidence level, locally without the cloud
Atmos	Object-based 3D audio format from Dolby, recognized by MediaInfo
AV1	Modern, royalty-free video codec with high efficiency, successor to VP9
AVC	Advanced Video Coding, identical to H.264
Bit depth	Number of bits per audio sample or color channel, typically 16 or 24 bits for audio; higher values mean a finer resolution
Bitrate	Amount of data per unit of time, typically in kilobits or megabits per second, often an indicator of the quality of a video or audio stream
Bitrate EquivalenceLab	Tool in the Media Lab that converts bitrates between codecs: which bitrate in a more modern codec matches a given bitrate in an older one at comparable quality
Blu-ray	Optical storage medium for high-definition video, typically uses MPEG transport streams
Broadcast	The broadcasting field, covering television and radio distribution with specific container formats such as MXF and transport streams
Chroma subsampling	Technique for reducing color resolution, common variants are 4:4:4, 4:2:2 and 4:2:0
Classification	Assignment of a check result to a defined category, see Appendix B
Codec	Set of rules for encoding and decoding video or audio data, for example H.264, H.265, AAC
CodecLab	Tool in the Media Lab that classifies codecs and their properties and helps you choose the right format for a given purpose
Color space	Mathematical description of the colors that can be displayed, common spaces are BT.709 (HD) and BT.2020 (UHD)
Compatibility assessment	Rule-based estimate of playability based on technical media data

Term	Explanation
Container	File format that bundles video, audio and metadata streams in one file, for example MP4, MKV, AVI
CRC	Checksum (Cyclic Redundancy Check), used to detect data errors
Decoding	Converting an encoded media file into displayable images and sound
DeepCheck	In-depth type integrity check with the Magika plug-in
Diagnostic	Check mode with a three-stage in-depth analysis (stream copy test, ffprobe in-depth analysis, decode layer) for stubborn cases; the result appears in the diagnostic report
DTS	Digital Theater Systems, an audio codec, common on Blu-ray and DVD
DTS:X	Object-based 3D audio format from DTS, recognized by MediaInfo
EBML	Extensible Binary Meta Language, the structural foundation of the Matroska container format
Emby	Media management system, reads NFO files for metadata
FFmpeg	Plug-in for processing and decoding media files
FFprobe	Plug-in from the FFmpeg package for reading metadata
FLAC	Lossless audio codec (Free Lossless Audio Codec)
Frame	Single image of a video stream
Frame rate	Number of frames per second, typical values: 24, 25, 30, 50, 60 fps
FREE	Free license tier for private, non-commercial use
Full check	Complete decoding of the entire file for an in-depth integrity check
H.264	Widely used video codec, also called AVC, the standard for HD video content
H.265	Successor to H.264, also called HEVC, more efficient at high resolutions
Hash value	Fixed character string calculated from a file for integrity checking, here SHA-256
HDR	High Dynamic Range, an extended dynamic range for bright and dark image areas
HEVC	High Efficiency Video Coding, identical to H.265
Image format	Classification of the aspect ratio into five shapes: Vertical, Square, Classic, Widescreen, Cinemascope
Integrity check	Check of the file structure and decodability, here based on a 30-second sample
ISO BMFF	ISO Base Media File Format, the foundation of MP4 and MOV
Jellyfin	Free media management system, reads NFO files for metadata

Term	Explanation
JSON	Text-based data format for the structured storage of settings and metadata
Kodi	Widely used media management system, reads NFO files for metadata
LIFETIME	License tier with the PRO scope for all major versions, including future ones
Live telemetry	Display of total system CPU, VidiVerify CPU, memory, network and read rate in the main window
Magic bytes	Characteristic byte sequence at the start of a file that reveals the file format
Magika	Plug-in from Google for content-based file type detection
MAP	Media Analysis Protocol, VidiVerify's proprietary module for user-friendly presentation
MAP export	Saving the MAP as a single-page A4 PDF archive sheet via the “MAP” card on the “Media Analysis” tab
MAP+	Detailed expert report of the MAP, PRO feature in preparation
Matroska	Open, flexible container, also known as MKV
Media Batch	Tab for checking entire folders and collections in one run, PRO feature
Media Compare	Planned PRO module that compares two versions of the same file and recommends the better one
Media Lab	Dedicated tab serving as the launch pad for advanced, in-depth analysis tools: compatibility assessment and the tools CodecLab, Bitrate EquivalenceLab and Video SizeLab
Media Repair	Tab for finding-based repair of damaged files with mandatory verification, PRO feature
Media Test	Detail window for audio and video, opened from the “Media Analysis” tab (level, spectrum, bitrate, waveform, detection of dropouts and inflated bitrates)
MediaInfo	Plug-in for in-depth analysis of media files and stream parameters, also recognizes Atmos and DTS:X
Metadata	Descriptive information about a file, such as resolution, duration or codec
MKV	File extension for Matroska video containers
MKVToolNix	Tool collection for Matroska containers; VidiVerify uses its mkvmerge for Media Repair
MOV	Container format from Apple, technically related to MP4
MP3	Lossy audio codec, widely used
MP4	Common container format based on ISO BMFF
MPEG-2	Video compression standard, the foundation of DVD and digital television

Term	Explanation
MPEG-TS	MPEG transport stream, used in broadcasting and on Blu-ray
MXF	Material Exchange Format, professional container format for broadcast and production
NFO	Text-based metadata format for media management systems such as Kodi, Jellyfin and Emby
OpenCV	Library for image processing and computer vision, bundled in the VidiVerify Runtime
Opus	Modern, lossy audio codec with high quality
PCM	Pulse Code Modulation, uncompressed audio data
Perceptual quality analysis	Assessment of how a video is visually perceived, see VQA
Performance mode	Predefined profile for controlling CPU load: Gentle, Balanced, High-Performance
Precheck	Short automatic preliminary check when a file is loaded
PRO	Paid license with the full feature set for one major version, required for commercial use
Protocol (log)	Chronological record of events during operation of the application
QuickCheck	Check in seconds via right-click in Windows Explorer, without opening the main application
RepairMAP	Audit sheet of a repair with command line, tool version and SHA-256 checksums, PRO feature
RIFF	Resource Interchange File Format, the structural foundation of the AVI container
Rule set	The versioned collection of all of VidiVerify's rating rules, with an ID such as “VVRW26.13” in every report
Sample rate	Number of audio samples per second, measured in hertz (Hz), typical values: 44,100 Hz or 48,000 Hz
SHA-256	Cryptographic hash algorithm for checking file integrity
StapelMAP	Check protocol of an entire batch as a PDF, PRO feature
Stream	Single data flow within a media file, usually separated into video, audio or subtitles
Strict abort	Setting in which the very first decoding error ends the check
Subprocess	External program that VidiVerify starts and monitors, for example a plug-in
Support package	Automatically compiled package with diagnostic information for support
Timeout	Time limit after which a process is ended

Term	Explanation
Transport stream	Continuous data stream with fixed packet sizes, the foundation of digital television
Video SizeLab	Tool in the Media Lab that estimates the effect of scaling and distinguishes simple upscaling (UPS) from AI-assisted methods (VSR)
VidiVerify Runtime	Dedicated Qt runtime environment, loaded on demand, with bundled professional libraries (PyAV, OpenCV) for the advanced windows
VP9	Video codec, often used in the WebM container
VQA	Video Quality Assessment, the perceptual quality analysis
VV-ID (VidiVerify ID)	Stable, anonymized device identifier with the prefix VV-, used to match support cases; it allows no conclusions about the person or the hardware
VVProgressiv™	Planned PRO check that examines short sequences spread across the entire runtime and automatically goes deeper at conspicuous points
Watchdog	Monitoring component that checks the user interface for hung states
WebM	Open container format based on Matroska, usually with VP9 or AV1
YTUP™	YouTube upload forecast, planned PRO module of VidiVerify

Appendix B: Rating System and Status Classes

VidiVerify assigns every check a clear status class. This classification is part of the MAP (Media Analysis Protocol) and lets you place the result at a glance. It forms the basis for follow-up decisions such as archiving, further processing or making a new copy.

B.1 Status classes

Status	Meaning
OK	The check completed without errors. Within the selected check level, the file is considered unremarkable.
WARNING	The check ran through technically but found indications of anomalies. An assessment is recommended, for example with the diagnostic.
ERROR	The check encountered an error. The file has structural or decoding problems and should not be used further without checking.
NOT CHECKED	No result is available yet for this check level.

B.2 Rating levels

The rule set assigns technical values such as the bitrate to one of four levels. A uniform color family applies throughout the application, in every window and every report:

Level	Meaning
OK	The value meets the requirement of the rule set
sufficient	Usable, with noticeable compromises
low	Clearly below what is usual for this content
critical	So low that quality problems are likely

Some formats have their own thresholds, for example HE-AAC, which sounds considerably better than AAC-LC at the same bitrate. The “Rating details & symbols” window (Section 10.6) shows all rules and how they are derived.

B.3 Resolution classes

In addition, the MAP assigns the footage to a resolution class:

Class	Short or long side from	Typical use
SD	below that	Older recordings, DVD, early mobile devices
HD	720 or 1280	Web, streaming, broadcasting

Class	Short or long side from	Typical use
Full HD	1080 or 1920	Home cinema, professional production
2K	1440 or 2560	High-quality web and gaming recordings
2.7K	around 1520 at about 2700 on the long side	Action and drone cameras
4K UHD	2160 or 3840	Cinema, high-end streaming productions
8K	4320 or 7680	Specialized productions

Orientation does not matter: a portrait phone video is classified like landscape. Movies without black bars, such as 1920x800, count by their full width and are therefore Full HD.

B.4 Image format

VidiVerify classifies the aspect ratio everywhere with the same division into five shapes, each with its own symbol:

Image format	Aspect ratio	Example
Vertical	portrait	Smartphone videos, Reels, Stories
Square	about 1:1	Square social media formats
Classic	above 1.05:1 up to 1.5:1	4:3 television, older recordings
Widescreen	above 1.5:1 up to 2.3:1	16:9, most of today's videos
Cinemascope	above 2.3:1	Cinema formats such as 2.39:1

B.5 Classification notes

In addition to the status class, VidiVerify states the cause in text form. Examples:

- Unremarkable container and stream structure
- Additional streams detected (for example subtitles, chapter markers)
- Timeout during the check
- Network source not stable during the check
- Faulty file or stream access
- No streams detected
- No valid container format detected: not repairable, check the file or obtain it again
- Plug-in error

> **[i] Purpose of the classification**
>
> The status class delivers a fast, color-coded statement about the result. The text classification adds the cause, so that even with errors it is immediately clear where the problem comes from. Together, both are a core feature of the MAP.

Appendix C: Supported File Formats

VidiVerify supports a wide range of common video and audio formats. Detection uses both the file extension and the file's header signature (magic bytes), so renamed files are also classified reliably.

C.1 Container formats

Container family	Typical extensions	Description
MP4/MOV (ISO BMFF)	mp4, m4v, m4a, mov, 3gp, 3g2	Widely used, streaming, mobile devices, professional production
Matroska/WebM (EBML)	mkv, webm, mka	Flexible container, widely used on the internet
AVI (RIFF)	avi	Classic Windows container format
MPEG program stream	mpg, mpeg, vob, mod, tod	DVD environment, older consumer devices
MPEG transport stream	ts, m2ts, mts	Broadcasting, Blu-ray, IPTV
Windows Media (ASF)	wmv, wma, asf	Older Windows-based productions
Flash Video	flv	Historical web content
MXF	mxf	Broadcast and professional production
Ogg	ogv, ogg	Open media formats
RealMedia	rm, rmvb	Historical streaming content

C.2 Notes on codecs

The actual decoding is handled by the FFmpeg plug-in. All codecs available in the installed FFmpeg version are supported. These include H.264, H.265, VP8, VP9, AV1, MPEG-2, MPEG-4, AAC, MP3, FLAC, Opus, PCM and many more.

C.3 Audio file formats

VidiVerify also checks pure audio files as a category of their own (see Section 6.12). The following audio file formats are recognized and handled:

Audio family	Typical extensions	Description
MPEG audio	mp3, mp2	Widely used, lossy audio formats
MP4 audio (AAC)	m4a, aac	AAC audio in the ISO BMFF container or as a raw stream

Audio family	Typical extensions	Description
FLAC	flac	Lossless compression at full quality
WAV / AIFF (PCM)	wav, wave, aif, aiff, aifc	Uncompressed PCM audio data in studio and archive quality
Ogg family	ogg, oga, opus, spx	Open formats with Vorbis, Opus or Speex audio
Windows Media Audio	wma	Audio format from the Windows Media environment
Matroska audio	mka	Audio track in the flexible Matroska container
Lossless (APE / WavPack)	ape, wv	Monkey's Audio and WavPack, lossless compression
Dolby / DTS	ac3, eac3, dts	Multichannel audio formats from Dolby and DTS, also as standalone files
AMR	amr	Speech-optimized format from mobile telephony

C.4 Audio codecs

The audio of the formats listed above, as well as the audio tracks in video containers, is decoded and analyzed using the following audio codecs:

Codec	Description
MP3 / MP2	MPEG audio, lossy, very widely used
AAC	Advanced Audio Coding, lossy, standard in MP4; HE-AAC v1 and v2 with their own rating thresholds
ALAC	Apple Lossless, lossless
FLAC	Free Lossless Audio Codec, lossless; the full check verifies the embedded MD5 checksum
PCM	Uncompressed audio data, for example in WAV and AIFF
Opus	Modern, lossy codec with high quality
Vorbis	Open, lossy codec in the Ogg environment
Speex	Speech-optimized, lossy codec
WMA	Windows Media Audio, lossy and lossless
Monkey's Audio (APE) / WavPack	Lossless compression formats
AC-3 / E-AC-3	Dolby Digital and Dolby Digital Plus, multichannel audio
DTS	Digital Theater Systems, multichannel audio, including DTS:X

Codec	Description
AMR	Adaptive Multi-Rate, speech compression from mobile telephony

[+] Regular updates recommended

New codecs and container variants are continuously integrated into FFmpeg. The automatic update check for the plug-ins ensures that newer formats are also reliably recognized and checked.

Appendix D: Access and Controls

You operate VidiVerify through its graphical interface. Dialogs can be controlled entirely by keyboard, the tab bar with the arrow keys. The overview summarizes the most important ways to access each function.

D.1 Main operation

Action	Access
Choose a file or folder	File selection in the main window (“Choose files”, “Choose folder”) or drag files into the window
Check from Explorer	Right-click: “Check with VidiVerify” (QuickCheck) or “Open with” > VidiVerify
Check level and check mode	Selection in the main window: Integrity check or Full check, Standard or Diagnostic
Start and cancel a check	Buttons in the main window
Run the diagnostic afterwards	“Start diagnostic” after a check with an anomaly
Open the diagnostic report	Diagnostic card on the Overview tab
Save the MAP	“MAP” card on the “Media Analysis” tab
View the rule set	“Rating details” card on the “Media Analysis” tab
Check a batch (PRO)	Choose several files or a folder, “Media Batch” tab
Resume a batch (PRO)	“Resume batch …” in the file selection or on the “Media Batch” tab
Repair a file (PRO)	“Media Repair” tab or “Start Media Repair” in the diagnostic report
Save the protocol	“Protocol” tab, “Save protocol” button
Settings	“Menu > Settings”
License	“Help > Licensing …”
Plug-ins	“Help > Plug-ins”
Release notes	“Help > Release Notes”
Exit the application	Close the window; if a process is running, VidiVerify asks for confirmation

D.2 Error handling and support

- Exceptions are written automatically to logs/exceptions.log
- If processes hang, VidiVerify saves a diagnostic dump in logs/hangs

- The VV-ID appears in the About window and the license window and helps with support requests
- The support package automatically compiles all relevant diagnostic information
- Support: support@vidiverify.de

D.3 Image credits

The screenshots show scenes and the poster of the film “Tears of Steel” (2012): (CC) Blender Foundation | mango.blender.org, licensed under Creative Commons Attribution 3.0 (CC BY 3.0).

> **[*] In closing**
> VidiVerify is a reliable companion for everything related to the quality of media files: check, rate, document and, where needed, repair. Carefully selected plug-ins, the proprietary MAP, an open rule set and clear operation come together in a tool that fits into existing workflows and keeps growing with every update.